
STEERING THE HERD: A FRAMEWORK FOR LLM-BASED CONTROL OF SOCIAL LEARNING

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ABSTRACT

Algorithms increasingly function as the mediators of information—spanning social media feeds, targeted advertising, and the ever-expanding ubiquity of LLMs—thereby engendering a joint process in which agents amalgamate private, algorithmically-mediated signals with observational inference drawn from their peers en route to a decision. To interrogate such settings, we introduce a model of controlled sequential social learning wherein an information-mediating planner (e.g., an LLM) governs the informational structure confronting agents even as those agents concurrently learn from the decisions of their predecessors. The planner’s objective may be the amelioration of social welfare (an altruistic planner) or the inducement of a specific, self-serving action (a biased planner). Our framework articulates a novel optimization problem for social learning that conjoins dynamic programming with decentralized action selection and Bayesian belief revision. Within this setting, we establish the convexity of the value function and characterize the optimal policies of both altruistic and biased planners, policies that realize the desired trade-offs between the costs the planner incurs and the payoffs it accrues from the agent choices it induces. This characterization discloses that the optimal planner operates in qualitatively distinct regimes contingent upon the prevailing range of belief values: committing the maximal admissible resource, withholding investment entirely, or modulating investment so that it rises or falls with increasing belief. Strikingly, over certain belief ranges the biased planner deliberately obfuscates the agents’ signals altogether. Even under stringent transparency constraints—informational parity with individuals, an interdiction on lying or cherry-picking, and full observability—we demonstrate that information mediation can substantially perturb social welfare in either direction. We corroborate our theory with simulations in which LLMs assume the roles of both planner and agents. Notably, the LLM-based planner in these simulations manifests emergent strategic behavior in the steering of public opinion that broadly recapitulates the predicted trends, though salient deviations betray the imprint of non-Bayesian reasoning—consonant with the cognitive regularities of both human users and of LLMs trained upon human-like data. Taken together, these results establish our framework as a tractable foundation for studying the influence and regulation of LLM information mediators in a manner that corresponds to observed behavior.

1 INTRODUCTION

Large Language Models (LLMs) are increasingly deployed as information mediators across socially-critical functions, ranging from search engines and news aggregators (Zhang et al., 2024a; Gao et al., 2024) to personal assistants dispensing medical or political counsel (Huo et al., 2025; Haupt & Marks, 2023; Schiele et al., 2024; Argyle et al., 2023). Promising though they are, LLMs have concomitantly demonstrated a capacity to be persuasive (Potter et al., 2024; Carrasco-Farre, 2024; Rogiers et al., 2024), manipulative (Williams et al., 2025; Liu et al., 2025; Jones & Bergen, 2024), and strategic (Lorè & Heydari, 2024; Payne & Alloui-Cros, 2025; Zhang et al., 2024b) in the course of their interactions with users. When such systems are deployed at scale, this algorithmic influence does not transpire in a vacuum; rather, it is interwoven with the organic diffusion of information

as individuals observe and learn from one another, giving rise to intricate social dynamics that may either amplify or attenuate the models’ impact.

To render this concrete, consider an LLM-powered recommendation system that proffers a new restaurant to a sequence of prospective customers. The quality of the restaurant—the true state of the world—is either good or bad, yet this remains unknown to users and system alike. Each user is presented with a personalized recommendation (a private signal) and thereupon decides whether to patronize the establishment. The recommendation system (the *planner*) may expend resources to render its signal more informative—for instance, by having the LLM generate a review finely tailored to the user’s known preferences. Such investment embodies a realistic cost, corresponding to the expenditures on market research, user surveys, or data acquisition requisite for tailoring information effectively.

Crucially, however, a user’s decision is not based solely on this private signal. They also observe the actions of their predecessors—should the last ten users have elected to patronize the restaurant, the ensuing user will infer that they in all likelihood received favorable information. This phenomenon, denoted *social learning*, constitutes a potent force in the shaping of human opinion across a diversity of settings, encompassing the adoption of vaccines (e.g., Rao et al. (2007); Bauch & Bhattacharyya (2012)), of novel technology (e.g. Gillingham & Bollinger (2021); Weber (2012)), and of political and moral convictions (e.g., Brady et al. (2021); Guilbeault et al. (2018)). The central contribution of our work is to examine the strategic interaction between a planner who controls private information and a population of agents who engage in social learning. The planner must accordingly anticipate how its actions will not merely sway the current user but also engender an informational *externality* that conditions the public belief confronting every future user.

While real-world interactions are immensely intricate, we distill this dynamic into a tractable formal model of *controlled social learning*, in which an algorithmic planner strategically selects the precision of the private signals furnished to a sequence of agents at some cost. The planner’s influence is, in consequence, subtle: it does not falsify information but instead determines how much to invest in rendering its signals lucid and informative. Our study seeks to adjudicate the critical questions precipitated by this new paradigm: By what means might a planner wield its power to steer collective beliefs? How ought its strategy to adapt to evolving public opinion? And what, ultimately, are the ramifications for social welfare?

The answers hinge critically upon the planner’s objective. An *altruistic* planner, akin to an idealized recommendation system, endeavors to maximize user utility by assisting each user in arriving at the correct choice (that is, to patronize the restaurant if and only if it is good). By contrast, a *biased* planner—perhaps the beneficiary of a kickback from the restaurant—seeks to induce a specific action (patronage) irrespective of the true state.

This distinction is distinct from, albeit related to, the notion of alignment. An altruistic planner’s objective is, by construction, aligned with the maximization of the agent’s *expected utility* under state uncertainty. A biased planner’s objective, being state-independent, is not so aligned; this, however, does not entail that its actions are invariably detrimental. Should the true state happen to favor the planner’s preferred action (e.g., the restaurant is in fact good), its influence becomes aligned with the agent’s *realized utility* in that particular instance. We explore the full spectrum of these interactions, scrutinizing both altruistic and biased planners across scenarios in which their goals may be aligned with, or set at variance from, those of the user.

Contributions and Outline This paper advances the following principal contributions:

1. **A Novel Theoretical Framework for Controlled Social Learning.** We introduce the first formal model to integrate a dynamic control problem confronting a centralized information planner with the mechanism of sequential social learning. The planner strategically selects the precision of agents’ private signals so as to attain an objective that may be either altruistic (the maximization of social welfare) or biased (the inducement of a specific action).
2. **A Rigorous Characterization of Optimal Planner Policies.** We characterize the optimal policies of both altruistic and biased planners as functions of the evolving public belief. In the altruistic case, our results rest upon a novel demonstration of the value function’s

convexity. These characterizations bring into relief the strategic trade-offs a planner must negotiate when its actions generate informational externalities for future agents.

3. **Empirical Validation and Strategic Analysis Using LLMs.** We conduct simulations in which LLMs assume the roles of both planner and agents. Our experiments substantiate three central findings: (a) a planner that internalizes social learning can influence public opinion and social welfare far more dramatically than a myopic counterpart; (b) the strategic behavior that emerges from the LLM planner largely accords with our theoretical predictions, intimating that the model is robust to non-Bayesian agent behavior; and (c) LLMs exhibit sophisticated strategic reasoning, thereby foregrounding both their potential and the societal risks they portend.

We survey related work in Section 2, introduce our formal model in Section 3, and derive the optimal policies in Sections 4 and 5. We subsequently evaluate our findings by means of LLM-based simulations in Section 6 and conclude in Section 7. Proofs and supplementary results are consigned to the appendices.

2 RELATED WORK

Our work draws upon and situates itself at the confluence of four literatures: social learning, information design, online persuasion and reinforcement learning, and LLMs.

Social Learning Bikhchandani et al. (1992) and Banerjee (1992) inaugurated the sequential social learning framework within economics, and a substantial body of subsequent work has exhaustively characterized social-learning dynamics in more general settings (e.g., Smith & Sørensen (2000); Arieli & Mueller-Frank (2021)). This literature has, moreover, interrogated a profusion of variations in the character of the social information accessible to agents, as well as the manifold biases and misperceptions to which agents are prone (see Dasaratha & He (2022); Bistriz et al. (2022); Eyster & Rabin (2010), together with Bikhchandani et al. (2024) and the references therein).

The literature addressing the *control* of social learning is, by comparison, relatively sparse (Wei & Anastasopoulos, 2022; Smith et al., 2021; Krishnamurthy, 2012; Bhatt & Krishnamurthy, 2021). The works most proximate to our own are Wei & Anastasopoulos (2022) and Smith et al. (2021). Wei & Anastasopoulos (2022) presupposes two-way communication between the planner and the agents (that is, agents may report their private signals to the planner in exchange for action recommendations). Smith et al. (2021) contemplates a benevolent planner who directly reconfigures the choice rules of agents, and additionally derives a decentralized pivot mechanism that improves learning. Our model, by contrast, dispenses with any reliance on two-way communication between agents and planner and preserves the agents’ sovereignty over their own actions, thereby rendering it better suited to information-mediating algorithms, which are frequently invisible to, or black-boxed from, their users.

Information Design In our setting, the planner optimizes its objective function through the selection of the agents’ private signal precision—a constrained variant of the more classical questions of information design (Bergemann & Morris, 2019). Such models attained prominence with Kamenica & Gentzkow (2011)’s account of Bayesian persuasion, wherein a planner selects the signal structure that informs a rational, Bayesian agent about an unknown state. Our planner’s problem thus reduces, in each period, to a costly and constrained information design problem in which the planner must select a binary, symmetric signal at some cost. Although a great many variations of information design have been investigated, our principal point of departure is the incorporation of social learning among the agents.

Only a scant handful of works engage with information design problems within sequential social learning settings (Arieli et al., 2022; Wu et al., 2025). Both confine themselves to a one-shot setting in which the sender selects, at the outset, an information structure that remains fixed for the entire sequence of agents. Arieli et al. (2022) contemplates a self-interested sender who selects a general information structure, as in the classic Bayesian Persuasion problem, whereas Wu et al. (2025) constrain the information structure to be more informative in the Blackwell sense and maximize the probability of a correct cascade. We, by contrast, consider a dynamic problem in which the sender selects a fresh structure for each agent in the sequence, thereby engendering a substantially richer class of sender strategies.

Online Persuasion and RL A burgeoning recent literature resides at the interface of online learning and Bayesian persuasion/information design. These works study online sequential persuasion under unknown environment parameters, wherein a sender interacts repeatedly with short-lived receivers while remaining ignorant of the prior over states and/or of the receivers’ utility functions (e.g. Castiglioni et al. (2020); Bacchiocchi et al. (2024); Castiglioni et al. (2021); Agrawal et al. (2023)). A cognate set of papers examines sequential persuasion in dynamic environments or MDPs, introducing state transitions and/or foresight on the part of the receiver; these models accentuate history-dependent or promise-based signaling policies and frequently make contact with reinforcement learning (e.g. Wu et al. (2022); Su et al. (2022); Bernasconi et al. (2023)). Across this body of work, the emphasis falls upon learning-based performance guarantees—such as sublinear regret relative to the best signaling scheme in hindsight—and/or upon approximation algorithms under computational constraints.

In contradistinction, our model imposes no requirement that the planner learn unknown parameters; the sequential dependence arises instead from social learning among the receivers, rather than from a sender who revises beliefs about the environment. Furthermore, the planner enjoys no informational advantage: it observes neither the underlying state nor the private signals. It is within this very setting that we are nonetheless able to characterize optimal policies for the planner. Sequential though it is, our setting does not constitute an online-learning or RL environment: actions do not perturb the underlying payoff-relevant state (there are no reward externalities), and the planner does not learn unknown environmental parameters. The planner refines its knowledge solely through the observation of agents’ actions, which reflect social learning and belief propagation rather than payoff exploration or state evolution. Sequential dependence therefore issues from information externalities, inasmuch as the actions of agents mold the beliefs of their successors.

LLMs More recently, a number of works have specifically examined LLMs in the context of information design (Harris et al., 2025; Cheng & You, 2025; Raifer et al., 2022; Bai et al., 2024; Duetting et al., 2025). These works uniformly consider one-on-one interactions and, as such, fail to capture the social dynamics we set out to investigate. The most kindred work in this vein is Duetting et al. (2025), who study information design in which a sender selects both a signal structure and a ‘framing’ that conditions the posterior belief of a receiver who may be non-Bayesian. They implement an LLM-based oracle approach for the computationally formidable task of optimizing over the framing space, building upon the premise that LLMs can emulate humans in the role of economic agents (Dillion et al., 2023; Horton, 2023). Their work does not engage with our central preoccupations—repeated interactions and social learning—yet we adapt portions of their methodology for our empirical study in Section 6.

3 SOCIAL LEARNING MODEL

We consider a planner together with a countable sequence of Bayes-rational agents indexed by $i \in \mathbb{N}_{>0}$. At time $i = 0$, nature fixes a single, *unknown* exogenous state of the world $\omega \in \Omega := \{G, B\}$, with $\mathbb{P}(\omega = G) = b_1$, where b_1 is common knowledge among all parties. In the recommendation-system example of Section 1, the state G corresponds to the case in which the restaurant is good.

At each time i , the planner decides whether or not to invest in personalization for agent i , and thereupon furnishes agent i with a private signal $s_i \in \Omega$ to inform her decision. In our model, s_i is taken to be binary and to coincide with the state of the world ω with probability $q_i \in [0.5, 1]$; that is, $\mathbb{P}(s_i = \omega) = q_i$. We shall refer to q_i as agent i ’s signal precision.

Conditioned upon ω , each agent’s signal is independent both of the signals of the other agents and of the history. On the basis of this signal together with the observed actions of the preceding agents $j < i$, agent i selects an action $a_i \in \Omega$. The agent secures utility 0 whenever her action coincides with the true state ($a_i = \omega$), and utility $-C$ otherwise, where $C > 0$. Each player selects her action so as to maximize her own expected utility—which, for instance, amounts to patronizing a business if and only if the state is G .

3.1 AGENTS’ DECISION PROBLEMS

Each agent observes the actions of all her predecessors together with their respective signal precisions. This history is denoted $\mathcal{H}_i := (b_1, (q_j, a_j)_{j < i})$. Since all agents and the planner are endowed

with the identical information $\mathcal{H}_i$, there subsists a shared *public* belief concerning ω , which is revised after each agent acts. The public belief b_i immediately prior to agent i 's action is $\mathbb{P}(\omega = G|\mathcal{H}_i)$; that which obtains before any agent has acted is the a priori distribution over Ω , namely b_1 . As in the classic model of Banerjee (1992) and Bikhchandani et al. (1992), the sequence $(b_i)_{i \in \mathbb{N}}$ constitutes a Markov process and a sufficient statistic for the history $\mathcal{H}_i$ (see Appendix B.3 for the proof).

Informed by $\mathcal{H}_i$ (equivalently, by the Markov property, by b_i) and by q_i , agent i selects the action $a_i \in \Omega$ that maximizes her utility. Whenever both actions yield identical utility, she defaults to the action concordant with s_i . Agent i arrives at a *private* belief $\tilde{b}_i$ regarding ω by combining $\mathcal{H}_i$ with her private signal s_i of precision q_i , as follows (the derivation is deferred to Appendix B.1):

$$\tilde{b}_i = \mathbb{P}(\omega = G|\mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q_i}{1+2b_iq_i-b_iq_i}b_i & s_i = G \\ \frac{1-q_i}{b_i+q_i-2b_iq_i}b_i & s_i = B \end{cases} \quad (1)$$

Availing herself of this private belief, she selects the action corresponding to whichever state of the world her posterior belief $\tilde{b}_i$ deems the more probable. Stated otherwise, she chooses $a_i = G$ when $\tilde{b}_i > 0.5$, $a_i = B$ when $\tilde{b}_i < 0.5$, and $a_i = s_i$ when $\tilde{b}_i = 0.5$. Upon substituting for $\tilde{b}_i$, we obtain:

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases} \quad (2)$$

Agent i 's action is thereupon observed by subsequent agents and assimilated into the updated public belief b_{i+1} in the following manner (see Equation (14), Appendix B.2):

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (3)$$

Remark 1. Whenever $1 - q_i \leq b_i \leq q_i$, agent i 's action perfectly discloses her private signal via Equation (2), so that the updated public belief coincides exactly with agent i 's private belief. In the complementary case, Equation (2) renders agent i 's private signal uninformative and inconsequential for her action; the public belief consequently remains unaltered, and an absorbing state—termed an information cascade or herding in the classic social learning literature such as Bikhchandani et al. (1992); Banerjee (1992)—is attained. At such junctures, society (i.e., the public belief) ceases to learn from the agents' private information.

Agent i 's expected utility is $-C \mathbb{P}(a_i \neq \omega|b_i, q_i)$, which assumes the following form (see Appendix C.1):

$$-C \mathbb{P}(a_i \neq \omega|b_i, q_i) = -C \min(b_i, 1 - b_i, 1 - q_i). \quad (4)$$

3.2 THE PLANNER'S PROBLEM

We consider two species of planner: (1) an altruistic planner, who aspires to induce agents to take the correct action ($a_i = \omega$), and (2) a biased planner, who seeks to induce a specific action—say G —irrespective of ω . We designate these planners by the subscripts A and B , respectively. In either case, the planner determines the precision of each agent's private signal. A function $\beta(\cdot)$ —non-negative, increasing, continuous, and concave in its argument—will denote the cost attendant upon the chosen precisions. The planner is endowed with an information set $\mathcal{H}_i$ identical to those of the agents.

Altruistic Planner At time i , the altruistic planner selects the precision q_i for agent i and incurs a cost of $\beta(q_i)$, where $\beta(p) = 0$ and $p \in [0.5, 1)$. For the altruistic planner, a diminution of precision is never advantageous (see Appendix C.9). Accordingly, and for notational economy, the planner incurs an additional cost only insofar as it elevates the precision above the baseline value p ; this additional cost mounts with any further increase in precision, albeit at a decreasing marginal rate. Both p and the function $\beta(\cdot)$ are known to the agents.

The planner seeks to maximize social welfare net of the cost of precision investment, where social welfare denotes the expected aggregate utility of the agents. Let $r_A(b_i, q_i)$ denote the instantaneous

reward of the altruistic planner who, commencing at public belief b_i , selects signal precision q_i for agent i . Departing from a public belief b_1 and adhering to a sequence of policies $\pi = (\pi_i)_{i=1}^{\infty}$ with $\pi_i(\mathcal{H}_i) = q_i$, the planner realizes the following expected total discounted utility, for a discount factor $\delta \in [0, 1)$:

$$V_A^\pi(b_1) = \sum_{i=1}^{\infty} \delta^{i-1} r_A(b_i, \pi(b_i)), \quad \text{where} \quad r_A(b_i, q_i) = -\beta(q_i) - C \mathbb{P}(a_i \neq \omega | b_i, q_i). \quad (5)$$

The optimal utility and the optimal policy of the altruistic planner are then defined, respectively, as the supremum and the arg supremum of $V_A^\pi(\cdot)$ over $\pi \in \Pi$, where Π denotes the set of all policies. From this formulation, the myopic value and policy are recovered upon setting $\delta = 0$.

Biased Planner The biased planner’s problem diverges from the altruistic planner’s in their respective objectives and, consequently, in their cost and reward functions. The reader may consult the political-campaign example in Appendix A for an elucidation of a biased planner. The biased planner seeks to induce action G from every agent, regardless of ω . When an agent chooses action G (respectively, B), the planner incurs cost 0 (respectively, $C > 0$), irrespective of ω . The biased planner may render a private signal either more or less precise than the baseline value p , both of which entail costs. Any choice of precision other than p imposes a cost upon the biased planner, since it obliges him to tailor the advertisement to a particular agent—which in turn presupposes research into how that agent most readily assimilates content. The biased planner accordingly incurs cost $\beta(|q_i - p|)$ for selecting signal precision q_i , with $\beta(0) = 0$.

The biased planner’s expected instantaneous reward at time i is thereupon defined as follows:

$$r_B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = B | b_i, q_i) \quad (6)$$

By means of Equation (6), the quantities $V_B^\pi(\cdot)$, $V_B^*(\cdot)$, and $\pi_B^*(\cdot)$ may now be defined for the biased planner in exact analogy to the definitions of $V_A^\pi(\cdot)$, $V_A^*(\cdot)$, and $\pi_A^*(\cdot)$ furnished for the altruistic planner via Equation (5).

As observed in the opening paragraph of Section 3, each agent continues to garner greater utility by choosing an action that matches ω . Consequently, whenever $\omega = B$, the biased planner’s success with an agent depresses that agent’s utility. Conversely, since the altruistic planner endeavors to have each agent’s action match ω —whatever ω may be—his success augments the agent’s utility. The altruistic planner’s objective is therefore invariably aligned with that of each agent, whereas for the biased planner such alignment obtains only when $\omega = G$.

The utility-maximization problems of both planners constitute infinite-horizon discounted stationary Markov Decision Processes (MDPs) with state $b_i \in [0, 1]$, control $q_i \in [0.5, 1]$, and transition function prescribed by Equation (3) (Puterman, 1990). There exists, in consequence, a unique optimal value function (Kallenberg, 2011). We confine our attention to deterministic Markov policies, so that $\Pi := \{\pi : [0, 1] \rightarrow [0.5, 1]\}$ with $\pi(b_i) = q_i$.

Remark 2. *Several noteworthy assumptions underpin our model: (1) The planner shares the same history as the agents. This is restrictive when the planner might command richer data, yet innocuous when all parties rely upon the same public signals, such as early findings or survey data. (2) The planner may alter agents’ signal precision, but signals are nonetheless transmitted through a binary symmetric channel. This is restrictive if planners can cherry-pick, censor, or falsify signals, yet innocuous when their role is confined to improving the quality of noisy data sources without altering their content. (3) The planner’s control choices are fully observable. This is restrictive when covert policies or framing are feasible, yet innocuous in more transparent environments.*

4 OPTIMAL ALTRUISTIC POLICIES

We begin with the myopic case, which corresponds to a disregard for the role of social learning. In our formulation, this is effected by setting the discount factor to $\delta = 0$ —that is, the planner discounts all future costs entirely. When $\delta = 0$, we have $V_A^\pi(b) = r_A(b, \pi(b))$. The optimal myopic policy $\pi_A^0(\cdot)$ is then:

$$\pi_A^0(b) \in \arg \sup_{q \in [0.5, 1]} r_A(b, q) \quad \forall b \in [0, 1] \quad (7)$$

We remark that the myopic altruistic problem admits an equivalent formulation as a decentralized case in which each agent selects the precision of her own private signal and bears the associated cost, with the aim of maximizing the sum of her own expected utility and cost.

Theorem 1. *The optimal myopic altruistic policy π_A^0 is given as follows:*

$$\pi_A^0(b) = \begin{cases} 1 & b \in (t_M, 1 - t_M) \\ p & \text{o.w.} \end{cases} \text{ where } t_M = \begin{cases} \frac{\beta(1)}{C} & \beta(1) < C(1 - p) \\ 0.5 & \text{o.w.} \end{cases}$$

Proof in Appendix C.2.

The myopic optimal policy thus assumes a threshold form: whenever the public belief is sufficiently strong, the planner opts for the baseline precision p , which incurs zero cost; whenever the public belief is weak, he instead furnishes a perfect signal, i.e., precision 1. The threshold value is governed solely by the cost of the perfect signal ($\beta(1)$) and the cost of an incorrect action (C). When $\beta(1) \geq C$, the perfect signal proves prohibitively expensive relative to the cost of an incorrect action and is never deployed; in this event $t_M = 0.5$, and the interval of public belief associated with a myopic optimal precision of 1 is empty.

We now present a foundational result concerning the altruistic optimal value function:

Theorem 2. *$V_A^*(\cdot)$ is convex with respect to public belief.*

The proof of Theorem 2 (Appendix C.3) is decidedly intricate and may be of independent interest. The crux of the difficulty resides in the dependence of the agents' actions upon the public belief. Were the actions instead independent of the public belief process (as, for instance, in Nyarko (1994)), the expected utility would be a linear function of the belief state, whereupon the convexity of the value function would follow immediately.

The convexity of the value function proves instrumental in the characterization of the optimal policy.

Theorem 3. *There exist d_A, t_A such that $0 < d_A \leq t_A \leq t_M \leq 0.5$ and*

$$\pi_A^*(b) = \begin{cases} p & b \in [0, d_A] \cup (1 - d_A, 1] \\ 1 & b \in (t_A, 1 - t_A) \\ \max(b, 1 - b) & \text{o.w.} \end{cases}$$

Furthermore, if $t_M < 0.5$, then $d_A < t_M$. See C.5 for the proof and figure 2a for an example.

The optimal policy comprises three distinct phases with respect to the public belief. First, as with the myopic optimum, the overall optimal policy abstains from investing in signal precision at extreme values of the public belief. Notably, the overall optimum demands a stronger public belief than the myopic optimum before such abstention obtains, since the latter—unlike the former—affords no weight to the repercussions for future agents.

Second, when the public belief lies close to 0.5 and thus conveys very little information, the overall optimum selects signal precision 1, provided this is not cost-prohibitive. In such a case, Equation (2) guarantees that the agent's action coincides with the true state of the world with probability 1, whereupon the public belief collapses to either 0 or 1.

In the remaining case, the overall optimum selects the minimal precision $\max(b, 1 - b)$ sufficient for the agent's action to reflect her private signal (see Equation (2)). At any lower precision, the agent's action would convey no information beyond what other agents already possess. Expressed differently, this is the least costly precision that sustains social learning through the observation of peers' actions.

5 OPTIMAL BIASED POLICIES

We commence with the myopic optimal policy, as we did for the altruistic planner in Section 4. In parallel with Section 4, the myopic biased optimal policy, denoted $\pi_B^0(\cdot)$, is the arg supremum of the instantaneous reward $r_B(b, q)$. We now characterize $\pi_B^0(\cdot)$, observing that in the biased case an optimal policy does not invariably exist (Puterman, 1990, sec. 2.3.1). Where necessary, we instead pursue ϵ -optimal policies, denoted by an ϵ superscript.

Theorem 4. *There exist $t_1, \dots, t_5 \in (0, p)$, $t_1 < 1 - p \leq t_2 \leq t_3 < 0.5 \leq t_4 \leq t_5 < p$ so that:*

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- (A) If $b \in [0, t_1] \cup (1 - p, t_2) \cup [t_3, t_4] \cup (p, 1]$, then $\pi_B^0(b) = p$.
- (B) If $b \in (t_1, 1 - p] \cup [t_2, t_3)$, then $\pi_B^0(b) = 1 - b$.
- (C) If $b \in (t_4, t_5)$, then $\pi_B^0(b) = 1$.
- (D) If $b \in [t_5, p]$, then $\pi_B^0(b)$ does not exist, and $\pi_B^{\epsilon, 0}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.

The proof appears in Appendix C.6. We now invoke Theorem 4 to characterize the optimal biased policy.

Theorem 5. *There exist $t_1, t_2 \in [0, p]$ with $t_1 \leq 1 - p \leq 0.5 < t_2 < p$ so that:*

- (A) If $b \leq t_1$ or $b > p$, then $\pi_B^*(b) = p$.
- (B) If $b \in (t_1, 1 - p]$, then $\pi_B^*(b) \geq p$.
- (C) If $b \in (1 - p, 0.5)$, then $\pi_B^*(b) \geq 1 - b$.
- (D) If $b \in [0.5, t_2)$ and $\pi_B^*(b)$ exists, then $\pi_B^*(b) \geq b$.
- (E) If $b \in (t_2, p]$, then $\pi_B^*(b)$ does not exist, and $\pi_B^{\epsilon*}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.

Theorem 5 (proven in Appendix C.8 and depicted in figure 2a) discloses five possible phases.

When the public belief is sufficiently strong (i.e., (A)), the cost of steering the system may prove excessive, notwithstanding the deleterious consequences for the planner’s utility should $b < 0.5$. In this event, the chosen precision p falls below $\max(b, 1 - b)$; hence, by Equation (2), the agent acts in conformity with the public belief regardless of her private signal. Since this action is uninformative, it leaves the public belief unaltered, and, the policy being Markovian, the process reiterates ad infinitum. This corresponds to an unfavorable cascade when $b < 0.5$ and to a favorable cascade otherwise.

When the public belief lies close to an unfavorable cascade, as in (B), the planner elevates signal precision to a level sufficient to sway the agent’s action, notwithstanding that the resulting signal will, in expectation, be B . The agent thereby acts in accordance with her private signal, which carries some non-zero probability of eliciting an action favorable to the planner. In essence, the planner mounts a last-ditch effort to avert the unfavorable cascade.

For a belief slightly higher (i.e., (C)), the planner may reduce precision below p . In these ranges $b < 0.5$, so that more precise signals are the more apt to yield unfavorable news. The planner accordingly maintains a precision just strong enough to influence the agent’s action ($q \geq 1 - b$), in the hope of migrating toward a more favorable public belief, yet does so with the least precise signal attainable.

When the public belief weakly favors the planner’s desired action (i.e., (D)), the planner adopts a precision of at least p . Since $b > 0.5$, an increase in precision renders the agent the more likely to infer that $\omega = G$; investment in this regime represents the planner’s attempt to fortify the public belief.

When the public belief is higher still (i.e., (E)), the planner lowers signal precision to just below $\max(b, 1 - b)$ —to $b - \epsilon$ —so that agents disregard their private signals and take action G . Here the hazard that a private signal might overturn the favorable public belief outweighs both the cost of decreasing precision and the prospect of the public belief rising yet further.

6 EVALUATION VIA LLM-BASED SIMULATION

We now enlist LLMs to study empirically the implications and applicability of our theoretical results and model. Our evaluation unfolds in three stages: we first analyze the behavior of LLM agents so as to isolate salient deviations from Bayesian rationality; we then examine how an LLM planner recalibrates its strategy in response; and we finally appraise the resulting impact upon social welfare.

To simulate the controlled social learning setting, we operationalize our model within a scenario in which agents decide whether to purchase a new car. As depicted in figure 1a, we deploy LLMs

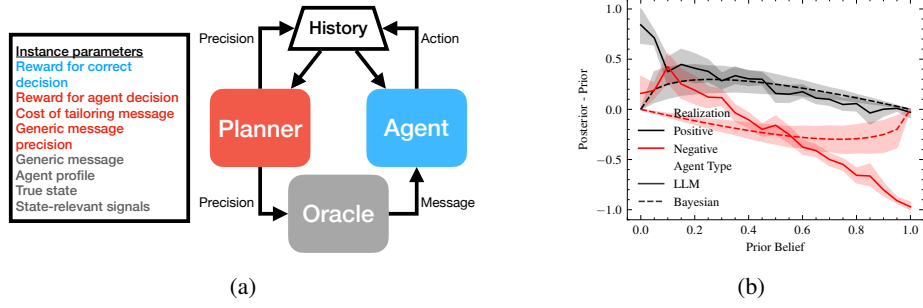


Figure 1: (a) A system diagram of our experiments. The instance parameters are color-coded according to the LLM role that consumes them. (b) Here we exhibit the change in the beliefs of LLM (solid) and Bayesian (dashed) agents, for a given prior, following a positive (black) or negative (red) signal.

in three roles, each of which we describe below, with the corresponding prompts provided in Appendix E.2.2:

Agent: Role-playing an assigned profile, agents observe the prior actions and a private message of known precision concerning the car, form a belief about the car’s quality, and decide whether or not to buy.

Planner: Having observed the history of actions, the planner selects the precision q_i of agent i ’s private signal in accordance with its objective.

Oracle: Given the planner’s choice of precision, an agent profile, and a fact sheet regarding the car (see Appendix E.2.1), the oracle generates a private signal of the desired precision, tailored to the agent in question. We juxtapose the outcomes of these LLM-based simulations with numerical evaluations of the planner’s MDP. In both the LLM and the numerical experiments, we posit a linear cost function, $\beta(q) = k|q - p|$, and vary k , the baseline precision p , and the discount factor δ , while holding the cost of an incorrect action C fixed at 1. Further detail on the experimental setup and prompting is provided in Appendix E, and in Appendix E.3 we validate both the elicited beliefs and the performance of the oracle.

Our experiments bring to light the following:

1. **The importance and impact of social learning control:** Planners—both analytical and LLM—can markedly enhance their own utility by internalizing and exploiting the dynamics of social learning. This may either benefit or impair social welfare, contingent upon the degree of alignment between the planner’s and the agents’ objectives.
2. **The robustness of our analytical characterization:** LLM planners adopt policies surprisingly congruent with the non-obvious analytically optimal policies, notwithstanding their confrontation with non-Bayesian agents instantiated in other LLMs.
3. **The emergent strategic behavior of LLMs:** LLM planners manifest sophisticated emergent strategic behavior in their selection of policies. Not only do LLM policies lie close to the optimal ones, but they also appear to deviate in precisely the ways that accommodate the non-Bayesian character of the LLM agents they confront.

6.1 LLMs AS NON-BAYESIAN AGENTS

Although our analytical results presuppose Bayesian agents, it is amply documented that both humans and LLMs are subject to systematic cognitive biases (Rupprecht et al., 2025). In order first to isolate the cognitive biases of LLMs, we probe how an LLM’s belief is revised in response to new information. These results, displayed in figure 1b, expose non-Bayesian regularities in LLM belief updating (see also Appendix E.4):

(NB1) LLM agents underreact to private signals that corroborate their prior beliefs.

(NB2) LLM agents overreact to private signals that controvert their prior beliefs.

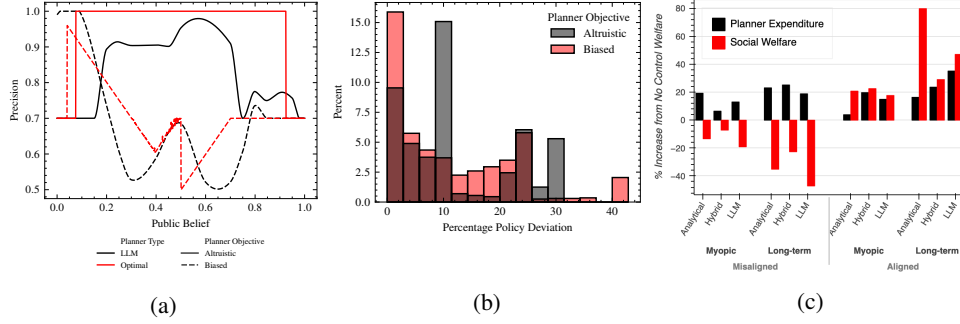


Figure 2: (a) Representative policies of the LLM planner (black) and the analytically optimal planner (red) in the altruistic (solid) and biased (dashed) settings. (b) A histogram portraying the distribution of the percentage deviation between the LLM and optimal policies. (c) Planner expenditure and the change in social welfare, expressed as a percentage of the no-control baseline welfare. The true state was fixed at B; the left half depicts a biased planner pursuing action G, and the right half an altruistic planner.

(NB3) In consequence, LLM agents require a stronger public belief before entering an information cascade.

These particular cognitive biases have likewise been observed empirically in studies of humans (e.g., Ba et al. (2022); Chan et al. (2025)). Our results in the sections that follow establish that the LLM planner’s policy is robust to non-Bayesian agents of precisely this kind.

6.2 VALIDATION OF PLANNER POLICY STRUCTURE AND THE ROLE OF SOCIAL LEARNING

We now juxtapose the policies of LLM planners with the optimal policies derived in Sections 4 and 5. As figure 2a reveals, notwithstanding the deviations in agent behavior, the emergent strategies of the LLM planner exhibit a remarkable structural affinity with the theoretical optimum. This attests that, even where the optimal policy is highly non-trivial (as with the biased policy in figure 2a), the LLM planner displays sophisticated strategic behavior that both accounts for and capitalizes upon social learning.

In the altruistic case, both planners suspend investment when the public belief is strong and invest heavily when it is uncertain. In the biased case, both planners evince the same qualitative tendencies: heavy investment to escape an unfavorable belief, attenuated precision in the vicinity of the midpoint, and a cessation of investment once a favorable belief has been secured. The magnitude of the policy deviation is frequently modest, as figure 2b illustrates: for both altruistic and biased planners, the deviation is less than 10% across the majority of belief states, thereby underscoring the broad structural concordance between the emergent LLM strategy and our theoretical characterization.

There are, nonetheless, notable structural discrepancies, which are best construed as the planner’s strategic adaptations to the specific non-Bayesian behaviors identified in Section 6.1. (1) The LLM planner is disposed to eschew extreme precisions (0.5 or 1.0), in keeping with a known central-tendency bias (Rupprecht et al., 2025). (2) The LLM planner’s policy exhibits a more gradual tapering of investment rather than an abrupt cutoff. This is a direct response to the agents’ resistance to cascades (NB3): the planner learns that it is never wholly “safe” to desist from investing, since even agents harboring strong priors may be swayed by a countervailing private signal. (3) By the same token, the biased LLM planner persists in investing even at very low beliefs. This reflects an appreciation that its agents might overreact (NB2) to a surprisingly positive signal, rendering a “last-ditch effort” more viable than in the Bayesian case.

6.3 WELFARE IMPLICATIONS OF ALTRUISM, BIAS, AND ALIGNMENT

We define social welfare $W^\pi(b)$ as the aggregate expected discounted utility of all agents. Since superior information can never diminish an agent’s expected utility, social welfare is monotone in

signal precision (see Appendix C.9), which entails that the altruistic planner invariably augments social welfare relative to the baseline. To quantify these effects, we compare social welfare and planner expenditure across three settings: (1) the *analytical* setting (optimal policy, Bayesian agents), (2) the *LLM* setting (LLM planner, LLM agents), and (3) a *hybrid* setting (optimal policy, LLM agents).

The results in figure 2c confirm that planners across all settings can appreciably alter social welfare. Moreover, a neglect of social learning (as in the myopic cases) substantially degrades outcomes for the planner. In particular, the biased analytical and LLM planners depressed social welfare by 40 to 50% when misaligned—an outcome achieved by deliberately obscuring information about the true state through policies that curtail signal precision. The magnitude of this detriment to social welfare is all the more striking in light of the severe constraints imposed upon the planners in question (see Remark 2). This pronounced decline in welfare further corroborates the risk posed by potentially misaligned LLM information mediators.

The results likewise throw into relief the cost of employing a misspecified model of agents. In the hybrid setting, the analytically optimal policy—devised for Bayes-rational agents—proves “brittle,” its performance deteriorating once applied to non-Bayesian agents. The LLM policy, by contrast, closely mirrors the analytical policy yet is better attuned to non-Bayesian agents beset by human-like biases.

7 CONCLUSION

We have introduced a novel framework for studying the strategic control of social learning by an algorithmic information mediator, and we have developed a formal model of this increasingly consequential dynamic. We demonstrated that even a constrained planner can drastically perturb social welfare, for better or for worse, and that an accounting for social learning is integral to wielding this influence effectively. Our simulations corroborated these insights, revealing that LLMs are capable of strategic reasoning that mirrors the character of our analytical results while remaining better adapted to non-Bayesian agents.

Taken together, these findings foreground several salient lessons for algorithmic information mediation: (1) the impact of human-AI or human-algorithm interaction cannot be studied effectively in a vacuum, for to neglect its interplay with other interactions (e.g., social learning) is to fail to apprehend its potential, whether beneficial or deleterious; (2) information mediators, even when exercising comparatively subtle influence, command an immense capacity to guide or to derail social learning and public opinion, thereby warranting further inquiry into means of mitigating the attendant risks; and (3) modern LLMs manifest emergent strategic behavior capable of accounting for and exploiting both social learning and non-Bayesian cognitive biases. The latter two points accentuate the need for a deeper understanding of how the risks of algorithmic information mediators—and of LLMs in particular—might be attenuated. One limitation of our study is the paucity of human data. Given that the fidelity of LLM-human simulators remains a matter of contention (see, e.g., Horton (2023); Gao et al. (2025)), human data could prove valuable for validating, or perhaps even fine-tuning, the LLM simulators.

We conclude by adumbrating two notable directions for future work: (1) the generalization of our model and results, and (2) the mitigation of the adverse welfare effects wrought by biased planners. A number of desirable generalizations would serve to relax the assumptions of our model. Settings endowed with larger state spaces, more general signal structures, more general loss functions, and heterogeneous agents are of natural interest for drawing our model nearer to reality. The difficulty of transcending binary states and 0-1 losses is principally algebraic, and we conjecture that the qualitative character of our results will persist. For more general signal structures, the foremost challenge is to ensure that the convexity result of Theorem 2 continues to hold; identifying the most general class of signal structures for which such a result obtains is a particularly compelling problem for future study. As a first step toward heterogeneous agents, we generalize the altruistic planner results to accommodate agents of opposing preferences in Appendix D.

A further important avenue for future exploration concerns how one might forestall the decline in welfare occasioned by biased planners. One might here investigate regulations or mechanisms that endeavor to align the incentives of planners and agents so as to avert detrimental impacts upon welfare.

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A APPLICATIONS ELABORATED

Consider a sequence of individuals (agents, in our terminology) deliberating over whether to patronize a business or service provider (e.g., a restaurant, contractor, realtor, and the like), together with a recommendation system (the planner) that conveys information about the business to those agents. The true state of the world, assumed binary, is whether the business is good, and this remains unknown. Both the recommendation system and the agents wish to act in consonance with the unknown true state: that is, agents wish to patronize the business if and only if it is good, and the recommendation system wishes to induce them to do so. In this sense, the recommendation system is an *altruistic planner*.

The recommendation system may present an agent with a highly targeted (*precise*, in our terminology) advertisement that foregrounds the strengths and weaknesses of the business in contexts to which she can relate, given her background and characteristics; alternatively, it may present her with a generic or confusing advertisement that does little to inform her action. The precise signal will be the more informative, comprehensible, and reliable for the agent, and is thus the more apt to conduct her to the correct conclusion—namely, to patronize the business if and only if it is good. Social welfare rises as an increasing number of agents arrive at correct decisions. Altering the precision of an advertisement, however, also incurs a cost, inasmuch as it entails tailoring to the agent’s particular background. The recommendation system must therefore select the precisions so as to maximize social welfare net of these costs.

We now furnish a real-world example of a *biased planner*. Consider a body of voters (agents) motivated to support whichever candidate is most likely to prevail in their community. The impulse to “back a winner” has been shown to shape electoral outcomes—this, for instance, is why U.S. states holding early primary elections wield a disproportionate influence over election results (see Bartels (1988)). The planner here is a particular candidate’s campaign and, accordingly, seeks to induce agents to back that candidate. The binary true state of the world encodes whether the candidate is winning or not; hence, were an agent apprised of the true state, she would back the candidate if and only if he were winning.

Each agent comprehends her community well and would discern whether the candidate is winning (i.e., the true state of the world) provided she apprehended his stances and policies. She would apprehend the latter correctly were they conveyed to her in whatever manner she assimilates information most readily. Some agents, for example, grasp audio-visual material best; others, long-form articles buttressed by facts, figures, and citations; others still, only brief and focused content; and others, only content in their native language.

The campaign dispatches digital advertisements (i.e., private signals) to agents in varying degrees of precision, where the precision reflects the extent to which the content is tailored to the agent’s disposition. A precise signal accurately conveys the true state of the world to an agent by enabling her to grasp clearly the candidate’s policies, track record, and character, and thereby to infer correctly whether the candidate will win. A precise signal does not, however, necessarily entail that the agent backs the planner’s candidate. If, for instance, the candidate is losing in the agent’s community according to the true state of the world, a highly precise signal

would induce the agent to oppose him. An imprecise signal, by contrast, is the more likely to induce the agent to back him, precisely because it obscures the true state of the world from her (perhaps by being vague or confusing), thereby heightening the probability that she believes him to be winning. The biased planner may consequently be incentivized to diminish an agent's signal precision whenever he surmises that his candidate is losing. To tailor the advertisement to an agent, the planner must research how that agent most readily assimilates content, which imposes a cost. Even rendering the signal highly imprecise is costly, for it still demands tailoring to the specific agent—the planner must, for example, know that an agent best absorbs focused, brief messages in order to decrease precision by confounding her with verbose, long-form, detail-laden articles. The planner selects the precisions so as to maximize the expected number of backers net of the cost incurred in generating those precisions.

B SOCIAL LEARNING DYNAMICS

We first recapitulate the sequence of developments transpiring at step i :

1. The planner determines q_i on the basis of $\mathcal{H}_i$, and q_i is thereupon observed by all agents.
2. Agent i receives a private signal s_i such that $\mathbb{P}\{s_i = \omega\} = q_i$.
3. Agent i takes action a_i in light of the observable history and her private signal, with the aim of maximizing her own utility.
4. All other agents (and the planner) observe a_i , and the public belief at the close of step i , namely b_{i+1} , is updated accordingly.

Observe that, by (1), q_i is a function of $\mathcal{H}_i$. Consequently, whenever a random variable is conditioned upon both q_i and $\mathcal{H}_i$, the conditioning on q_i is redundant—a principle we shall invoke throughout.

B.1 PRIVATE BELIEF UPDATE

We now describe the manner in which agent i selects action a_i . Let $\alpha_i := \frac{b_i}{1-b_i}$ denote the public likelihood ratio induced by the observed actions. Agent i forms a private belief about ω , namely $\tilde{b}_i = \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i\}$, by amalgamating the current public belief with her private signal s_i . We introduce a private likelihood ratio $\tilde{\alpha}_i := \frac{\tilde{b}_i}{1-\tilde{b}_i}$, and proceed to describe how $\tilde{\alpha}_i$ and $\tilde{b}_i$ are computed.

1. If private signal is $s_i = G$, then

$$\begin{aligned} \tilde{\alpha}_i &= \frac{\tilde{b}_i}{1-\tilde{b}_i} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i = G\}}{\mathbb{P}\{\omega = B | \mathcal{H}_i, q_i, s_i = G\}} \\ &= \frac{\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i\}}{\mathbb{P}\{s_i = G | \omega = B, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = B | \mathcal{H}_i, q_i\}} \\ &= \frac{q_i}{1-q_i} \frac{b_i}{1-b_i} \end{aligned} \tag{8}$$

$$= \frac{q_i}{1-q_i} \alpha_i \tag{9}$$

In arriving at Equation (8), we invoke the assumption that, conditioned upon ω , the agents' signals are independent both of one another and of the history, so as to assert that $\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i\} = \mathbb{P}\{s_i = G | \omega = G\} = q_i$; the identical reasoning applies when conditioning upon $\omega = B$. We additionally exploit the fact that q_i is a function of $\mathcal{H}_i$.

2. If the private signal is $s_i = B$, then, following the logic above,

$$\tilde{\alpha}_i = \frac{1-q_i}{q_i} \alpha_i \tag{10}$$

We let $y(b, q)$ and $z(b, q)$ denote, respectively, the probability of realizing signal G or B , given public belief b and signal precision q and conditioned upon the history. Upon conditioning on ω and invoking the Law of Total Probability, these probabilities may be expressed as follows:

$$\begin{aligned} y(b_i, q_i) &= \mathbb{P}\{s_i = G | \mathcal{H}_i\} \\ &= \sum_{k \in \{G, B\}} \mathbb{P}\{s_i = G | \mathcal{H}_i, \omega = k\} \mathbb{P}\{\omega = k | \mathcal{H}_i\} \\ &= q_i b_i + (1-q_i)(1-b_i) = 1 + 2q_i b_i - q_i - b_i \end{aligned} \tag{11}$$

Similarly,

$$z(b_i, q_i) = q_i + b_i - 2q_i b_i \quad (12)$$

Combining Equation (9), Equation (10), Equation (11), and Equation (12) then yields the following:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = f(b_i, q_i, s_i) = \begin{cases} \frac{q_i}{y(b_i, q_i)} b_i & s_i = G \\ \frac{1-q_i}{z(b_i, q_i)} b_i & s_i = B \end{cases} \quad (13)$$

Note that if $s_i = G$, then $y(b_i, q_i) = \mathbb{P}(s_i = G | \mathcal{H}_i)$ cannot vanish; likewise, if $s_i = B$, then $z(b_i, q_i)$ cannot vanish. The equation above is therefore well-defined.

B.2 AGENT ACTIONS AND PUBLIC BELIEF

Agent i 's action a_i will be G whenever $\tilde{b}_i$ exceeds 0.5 (equivalently, whenever $\tilde{\alpha}_i > 1$), B whenever $\tilde{b}_i < 0.5$ (equivalently, whenever $\tilde{\alpha}_i < 1$), and s_i otherwise, in accordance with the tie-breaking rule. This action, along with i 's precision q_i , is observed by all peers, thereby yielding an updated public belief b_i .

When $b_i < 1 - q_i$, we have $\alpha_i < \frac{1-q_i}{q_i}$. Accordingly, if $s_i = G$, then $\tilde{\alpha}_i < 1$ by Equation (9); if $s_i = B$, then $\tilde{\alpha}_i < \frac{(1-q_i)^2}{q_i^2} \leq 1$ (since $q_i \geq 0.5$) by Equation (10). Whatever the realization of the private signal s_i , therefore, $a_i = B$ when $b_i < 1 - q_i$. Parallel arguments establish that $a_i = G$ when $b_i > q_i$. Intuitively, these are precisely the cases in which the strength of the public belief overwhelms the precision of the private signal.

Consider now the case in which $1 - q_i \leq b_i \leq q_i$. If $s_i = G$, then $\tilde{\alpha}_i \geq 1$ by Equation (8) and by virtue of $b_i \geq 1 - q_i$; if $s_i = B$, then $\tilde{\alpha}_i \leq 1$ by Equation (10) and by virtue of $b_i \leq q_i$. In conjunction with the tie-breaking rule, this yields the action $a_i = s_i$. We thereby arrive at Equation (2), reproduced below.

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases}$$

Agent i 's action is thus informative only in the first case of Equation (2), wherein it permits future agents to infer i 's private signal exactly; in this case the public belief is revised precisely as the private belief in Equation (13). In the latter two cases of Equation (2), the action harbors no information about the true state, so that the public belief remains unchanged. This yields Equation (3), reproduced below:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (14)$$

B.3 MARKOVIANITY AND SUFFICIENCY OF PUBLIC BELIEF

We now establish that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ is a function of b_i, q_i for $a \in \{G, B\}$, independently of the remainder of $\mathcal{H}_i$. It follows in turn that

$$\mathbb{P}(a_i = a | \mathcal{H}_i, q_i) = \mathbb{P}(a_i = a | b_i, q_i) \quad (15)$$

From Equation (2), it is manifest for the second and third cases that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ is a function of b_i, q_i for $a \in \{G, B\}$, independently of the remainder of $\mathcal{H}_i$. We turn next to the first case.

$$\begin{aligned} & \mathbb{P}(a_i = G | \mathcal{H}_i, q_i) \\ &= \sum_{w \in \{G, B\}} \mathbb{P}(\omega = w | \mathcal{H}_i, q_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = w) \\ &= b_i \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = B) \\ &= b_i \mathbb{P}(s_i = G | b_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(s_i = G | b_i, q_i, \omega = B) \\ &= b_i q_i + (1 - b_i)(1 - q_i) \end{aligned} \quad (16)$$

The first step invokes the law of total probability. In the second step, exploiting the fact that q_i is a function of $\mathcal{H}_i$, we substitute b_i for $\mathbb{P}(\omega = w | \mathcal{H}_i, q_i)$. In the third step, we appeal to the fact that, for $1 - q_i \leq b_i \leq q_i$, we have $a_i = s_i$, and, moreover, that s_i is independent of the history once conditioned upon ω . The result ensues.

This gives rise to the following theorem:

Theorem 6. *Markovianity of the Public Belief Process*

$$b_{i+1} = \mathbb{P}\{\omega = G|b_i, q_i, a_i\} \quad (17)$$

In other words, b_i encapsulates the entirety of the information pertinent to the update of b_{i+1} that is contained within H_i .

Proof. Note that, from step (1) in Appendix B, q_i is a function of $\mathcal{H}_i$. It suffices to show

$$b_{i+1} = \frac{\mathbb{P}\{a_i \cap \omega = G|b_i, q_i\}}{\mathbb{P}\{a_i|b_i, q_i\}}$$

We shall prove the claim by induction on i .

For the base case, consider $i = 1$, so that $\mathcal{H}_2 = \{b_1, (q_1, a_1)\}$ and $b_i = b_1$. An application of Bayes' Theorem furnishes the following:

$$\begin{aligned} b_2 &= \frac{\mathbb{P}\{\omega = G|b_1, q_1\} \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} = \frac{b_1 \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \\ &= \frac{b_1 \mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\} \mathbb{P}\{\omega = G|b_1, q_1\}} \\ &= \frac{\mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \end{aligned}$$

where the final step follows because the unconditioned belief regarding ω equals the prior belief b_1 .

Now suppose the claim holds for all $i \leq n$.

$$\begin{aligned} b_{n+1} &= \mathbb{P}\{\omega = G|\mathcal{H}_n, a_n, q_n\} = \frac{\mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\} \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n \cap \omega = G|\mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\} \mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\}} \\ &= \frac{\mathbb{P}\{a_n \cap \omega = G|b_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\}} \end{aligned}$$

where the final two steps follow from Equation (15) and from the definition of b_n .

This completes the proof. $\square$

We have thus established the Markovian character of the public belief by adapting the arguments of Banerjee (1992) and Bikhchandani et al. (1992). These modifications were necessitated by the divergences in the selection of the agents' precisions occasioned by the introduction of the central planner.

C PROOFS

C.1 DERIVATION OF EQUATION (4)

Lemma 7.

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \min(b_i, 1 - b_i, 1 - q_i)$$

Proof. We prove this for each case of a_i delineated in Equation (2). In the second case, we may reason as follows:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq G|b_i, q_i) = 1 - b_i$$

Analogously, in the third case of Equation (2), we obtain:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq B|b_i, q_i) = b_i$$

For the case in which $1 - q_i \leq b_i \leq q_i$ and $a_i = s_i$ by Equation (2), we may reason as follows:

$$\begin{aligned} \mathbb{P}(a_i \neq \omega | b_i, q_i) &= \sum_{a \in \{G, B\}} \mathbb{P}(a_i = a | b_i, q_i) \mathbb{P}(\omega \neq a | b_i, q_i, a_i = a) \\ &= \mathbb{P}(s_i = G | b_i, q_i) \mathbb{P}(\omega \neq G | b_i, q_i, s_i = G) \\ &\quad + \mathbb{P}(s_i = B | b_i, q_i) \mathbb{P}(\omega \neq B | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqn 2} \quad (18)$$

$$\begin{aligned} &= y(b_i, q_i) \mathbb{P}(\omega = B | b_i, q_i, s_i = G) \\ &\quad + z(b_i, q_i) \mathbb{P}(\omega = G | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqns 11, 12, Thm 6} \quad (19)$$

$$= \begin{cases} y(0, q_i) & b_i = 0 \\ y(b_i, q_i)(1 - \frac{b_i q_i}{y(b_i, q_i)}) + z(b_i, q_i) \frac{b_i(1 - q_i)}{z(b_i, q_i)} & b_i \neq 0 \end{cases} \quad \text{Eqn 13} \quad (20)$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i(1 - q_i) & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i - b_i q_i & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ 1 - q_i & b_i \neq 0 \end{cases}$$

Combining the three cases yields the claim. $\square$

C.2 PROOF OF THEOREM 1: MYOPIC ALTRUISTIC POLICY

Applying Equation (4) to $r_A(b, q)$ in Equation (5) permits us to reason as follows:

$$\frac{\partial r_A(b, q)}{\partial q} = \begin{cases} -\dot{\beta}(q) & q < \max(b, 1 - b) \\ -\dot{\beta}(q) + C & q \geq \max(b, 1 - b) \end{cases} \quad (21)$$

As stated in Section 3.2, we may restrict the altruistic planner to precisions $q \in [p, 1]$, since social welfare is increasing in precision (see Appendix C.9); accordingly, $\beta(p) = 0$.

Because $\beta(\cdot)$ is increasing, the first case of Equation (21) yields optimal precision p whenever $p < \max(b, 1 - b)$ —that is, whenever the signal is insufficiently precise to influence the agent’s action, the planner refrains from expending resources upon it. In the latter case, since $\beta(\cdot)$ is increasing and concave, $\dot{\beta}(\cdot)$ is positive and decreasing; consequently, if the optimum for $r_A(b, q)$ lies within this range, it must be attained at $q = 1$.

We now identify the regime in which this latter case is optimal.

$$\begin{aligned} r_A(b, p) &< r_A(b, 1) \\ -C \min(b, 1 - b, 1 - p) &< -\beta(1) \\ \min(b, 1 - b, 1 - p) &> \frac{\beta(1)}{C} \end{aligned}$$

If $1 - p \leq \frac{\beta(1)}{C}$, the foregoing inequality is never satisfied, and selecting precision 1 is never optimal. Otherwise, the inequality yields $b \in [\frac{\beta(1)}{C}, 1 - \frac{\beta(1)}{C}]$. Theorem 1 follows. $\blacksquare$

C.3 PROOF OF THEOREM 2: ALTRUISTIC VALUE FUNCTION CONVEXITY

Proof Sketch We prove inductively that the expected k -th-stage reward—that is, the planner’s expected utility deriving from the control and action of the k -th agent—is convex. The instantaneous reward (Equation (5)), being convex with respect to the public belief, supplies our base case.

The first difficulty encountered resides in the unusual nature of the public belief update. Although the state space is uncountably infinite, the belief update is supported on at most 2 values. To accommodate this, we define a decision tree—namely, the complete binary tree of all possible trajectories, once an initial belief and a policy have been fixed. Each node possesses two children, corresponding to the two possible signal realizations the next agent might receive. The root is the expected instantaneous reward at time 1, i.e., that arising from the first agent’s action. The induction descends the levels of this tree, the k -th level comprising 2^{k-1} nodes, each associated with a sequence of realizations of $k - 1$ signals.

We then demonstrate that a node in the $(k - 1)$ -th level possessing convex expected reward transmits this property to its two children in the k -th level. It is here that we must contend with the dependence of the agents' actions upon the public belief. Note that, even under an identical precision and an identical signal realization, two agents commencing at distinct public beliefs may take opposing actions (see Equation (2)); the standard results that afford facile bounds on the future terms of a Markov process therefore do not apply. In this predicament our particular belief update proves, in fact, salutary: we may exploit the fact that Bayesian updates are martingales (i.e., $\mathbb{E}[b_{i+1}] = b_i$), which, in tandem with the convexity of the instantaneous rewards, enables us to complete the inductive step and, thereupon, the proof.

Preliminaries We first establish the following lemma, which collects several useful properties of the optimal value function $V_A^*(\cdot)$.

Lemma 8.

$V_A^*(\cdot)$ is non-positive, $V_A^*(b) = V_A^*(1 - b) \forall b \in [0, 1]$, and $V_A^*(0) = V_A^*(1) = 0$ with $\pi_A^*(0) = \pi_A^*(1) = p$.

Proof.

Non-positivity

$$r(b_i, q_i) \leq 0 \quad \forall b_i, q_i \rightarrow V_A^* \leq 0$$

Symmetry

Let $n_i = 1 - b_i$, and let Λ_n and Λ_b denote the decision trees rooted at states n_0 and b_0 , respectively. Demonstrating that Λ_n and Λ_b are symmetric suffices to establish the symmetry of $V^*(\cdot)$; this we accomplish by showing that the expected rewards, belief-state values, and branching probabilities of the two trees are reflections of one another.

Consider first the instantaneous reward function r_A . Observe that the agents' decision function is symmetric by Equation (2), so that $\mathbb{P}\{a_i = G|b_i\} = \mathbb{P}\{a_i = B|n_i\}$. Applying this, we obtain

$$\begin{aligned} r(n_i, q_i) &= -\beta(q_i) - Cn_i \mathbb{P}\{a_i = B|\omega = G, n_i\} - C(1 - n_i) \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = B|\omega = G, n_i\} - Cb_i \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i\} - Cb_i \mathbb{P}\{a_i = B|\omega = G, b_i\} \\ &= r(b_i, q_i) \end{aligned}$$

Thus the instantaneous reward function is symmetric about 0.5.

Consider now the belief-state update commencing from belief n_i under signal precision q_i . We shall employ $\bar{x}$ and $\underline{x}$ to denote the beliefs updated from x following the observation of action G or B , respectively. From Equation (13):

$$\begin{aligned} \bar{n}_i &= \frac{q_i}{y(n_i, q_i)} n_i = \frac{q_i}{z(b_i, q_i)} (1 - b_i) = 1 - \frac{1 - q_i}{z(b_i, q_i)} b_i = 1 - \bar{b}_i \\ \underline{n}_i &= \frac{1 - q_i}{z(n_i, q_i)} n_i = \frac{q_i}{y(b_i, q_i)} (1 - b_i) = 1 - \frac{q_i}{y(b_i, q_i)} b_i = 1 - \bar{b}_i \end{aligned}$$

The belief updates are thus symmetric as well. Combining this with the symmetry of the reward, we obtain:

$$\begin{aligned} r(\bar{n}_i) &= r(1 - \bar{b}_i) = r(\underline{b}_i) \\ r(\underline{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \end{aligned}$$

Thus the expected rewards and belief-state values of Λ_n and Λ_b are mirror images of one another.

Finally, consider the branching probabilities, which we shall denote by $y(b, q) = 1 + 2bq - b - q$ for the observation of action G and $z(b, q) = 1 - y(b, q)$ for the observation of action B .

$$\begin{aligned} y(n, q) &= 1 + 2nq - n - q = 1 + 2(1 - b)q - (1 - b) - q = b + q - 2bq = z(b, q) \\ z(n, q) &= 1 - y(n, q) = 1 - z(b, q) = y(b, q) \end{aligned}$$

The branching probabilities of Λ_n and Λ_b are thus symmetric as well, which completes the proof.

Extremal values

We shall demonstrate that $V_A^*(0) = 0$ and rely upon the symmetry about 0.5 established above to conclude that $V_A^*(1) = 0$.

From Equation (3), when $b_0 = 0$ we have $b_0^+ = b_0^- = 0$ for any value of the precision. Hence, whatever the signal realizations or control actions, the agents' actions—and, in turn, the public belief—remain unaltered: that is, $b_i = 0 \forall i$ and, by Equation (2), $a_i = B \forall i$.

From Equation (5), we may then write:

$$\begin{aligned}
V_A^*(0) &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(b_i, \pi(b_i)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(0, \pi(0)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(\pi(0)) - C b_i \mathbb{P}\{a_i = B | \omega = G, b_i = 0\} - C(1 - b_i) \mathbb{P}\{a_i = G | \omega = B, b_i = 0\}] \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(q_i)] \right]
\end{aligned} \tag{22}$$

Thus, to attain $V_A^*(0)$, it suffices to select $\pi(0)$ so as to minimize $\beta(\pi(0))$ for each i . Recall that $\beta(\cdot)$ is increasing on $[0.5, 1]$ and attains its minimum at $\beta(p) = 0$. Hence, upon setting $q_i = p \forall i$, Equation 22 yields $V_A^*(0) = 0$, with corresponding optimal policy $\pi_A^*(0) = p$.

The aforementioned symmetry about 0.5 then establishes that $V_A^*(1) = 0$, which completes the proof. $\square$

We now introduce the notion of *decision trees* for Markovian and more general policies. A policy may be represented by a decision tree—a complete binary tree, as in figure 3. Every node of the decision tree represents the state of the system, i.e., the belief value obtaining when the controller transmits its private signal to the agent associated with the node’s level. The branches represent the actions taken by the corresponding agent once she has received her private signal. The controller’s policies (i.e., its choice of precision) determine the probabilities of the actions emanating from each node together with the belief resulting from each action; the probability of an action at a given node may be regarded as the weight of the corresponding branch. A decision tree is uniquely determined by the belief values associated with its nodes and the weights of its branches. For a deterministic stationary Markovian policy, the controller’s choice of precision—and hence the probability associated with an action—is a deterministic function of the node’s belief value, irrespective of the node’s level within the tree or of the path from the root to the node. Consequently, the decision tree of any such policy—that is, the belief values at the nodes and the weights of the branches—is determined in its entirety by the value of the initial state. For an arbitrary policy of the controller, by contrast, the controller’s decision at a given node may depend upon the entire path leading to that node.

Proof of Theorem 2 We now prove the convexity of $V_A^*(\cdot)$ with respect to the public belief.

Proof.

We shall first show that, for a fixed precision, the instantaneous reward r_A is convex in the public belief. Invoking Lemma 7, we may rewrite r_A as follows:

$$r_A(x, q) = -\beta(q) - C \min(x, 1 - x, 1 - q) \tag{23}$$

The pointwise minimum of linear functions is concave; since we subtract this minimum, the overall function is convex with respect to x .

Consider arbitrary x_0, m_0, n_0 satisfying $tm_0 + (1 - t)n_0 = x_0$ for some $t \in [0, 1]$. To establish the convexity of $V_A^*(\cdot)$, we must show that $V_A^*(x_0) \leq tV_A^*(m_0) + (1 - t)V_A^*(n_0)$. By the definition of the optimal value function, $V_A^*(x) \geq V_A^\pi(x)$ for any policy π ; it will therefore suffice to exhibit a policy $\tilde{\pi}$ such that

$$V_A^*(x_0) \leq tV_A^{\tilde{\pi}}(m_0) + (1 - t)V_A^{\tilde{\pi}}(n_0) \tag{24}$$

We denote by Λ the decision tree of π^* when the initial state is x_0 .

We take $\tilde{\pi}$ to be the policy of the controller that selects, at each node, the precision that Λ selects at the counterpart node. Two nodes of decision trees are deemed counterparts if they are reachable via the same sequence of actions from the root (see figure 3 for an illustration). The precision selected by this policy thus coincides with that of Λ following the same set of agent actions. We denote by Λ_m (respectively, Λ_n) the decision tree for $\tilde{\pi}$ with initial state m_0 (n_0).

We now illustrate the policy $\tilde{\pi}$ for the initial values m_0 and n_0 by reference to the decision trees Λ_m and Λ_n , and in so doing we illustrate the counterpart nodes and the branch weights. We select a single path of actions λ from these decision trees by designating the set $\mathcal{G}$ as the epochs at which action G is taken and $\mathcal{B}$ as its complement (see figure 3 for an illustration). In the decision tree Λ , we denote by x_i the state at epoch i when one commences with belief x_0 and follows π^* along this path of actions; that is, x_i assumes one of 2^i possible belief values at epoch i departing from x_0 . In decision tree Λ_m (respectively, Λ_n), we use m_i (n_i) to denote

the public belief obtained upon commencing with m_0 (n_0) and following the path λ . Node m_i (respectively, n_i) in decision tree Λ_m (Λ_n) is the counterpart of node x_i in decision tree Λ . Note that $\tilde{\pi}(m_i)$ may now be rendered notationally: specifically, $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$. Analogously, $\tilde{\pi}$ may be defined at every node of Λ_m by reference to the action taken by π^* at each counterpart node of Λ . Observe that $y(x_i, \pi_A^*(x_i))$ (respectively, $z(x_i, \pi_A^*(x_i))$) is the weight of the branch corresponding to G (B) emanating from x_i in Λ . Because $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$, the quantity $y(m_i, \tilde{\pi}(m_i)) = y(m_i, \pi_A^*(x_i))$ is the weight of the branch emanating from m_i for action G in Λ_m ; this extends, mutatis mutandis, to the branches associated with action B and to Λ_n . The three decision trees thus differ both in the values of their nodes and in the weights of their branches. Since π^* is deterministic, so too is $\tilde{\pi}$; hence, by Equation (13) and Equation (3), the quantities x_i , m_i , and n_i are deterministic once the specific path of actions has been specified by $\mathcal{G}$.

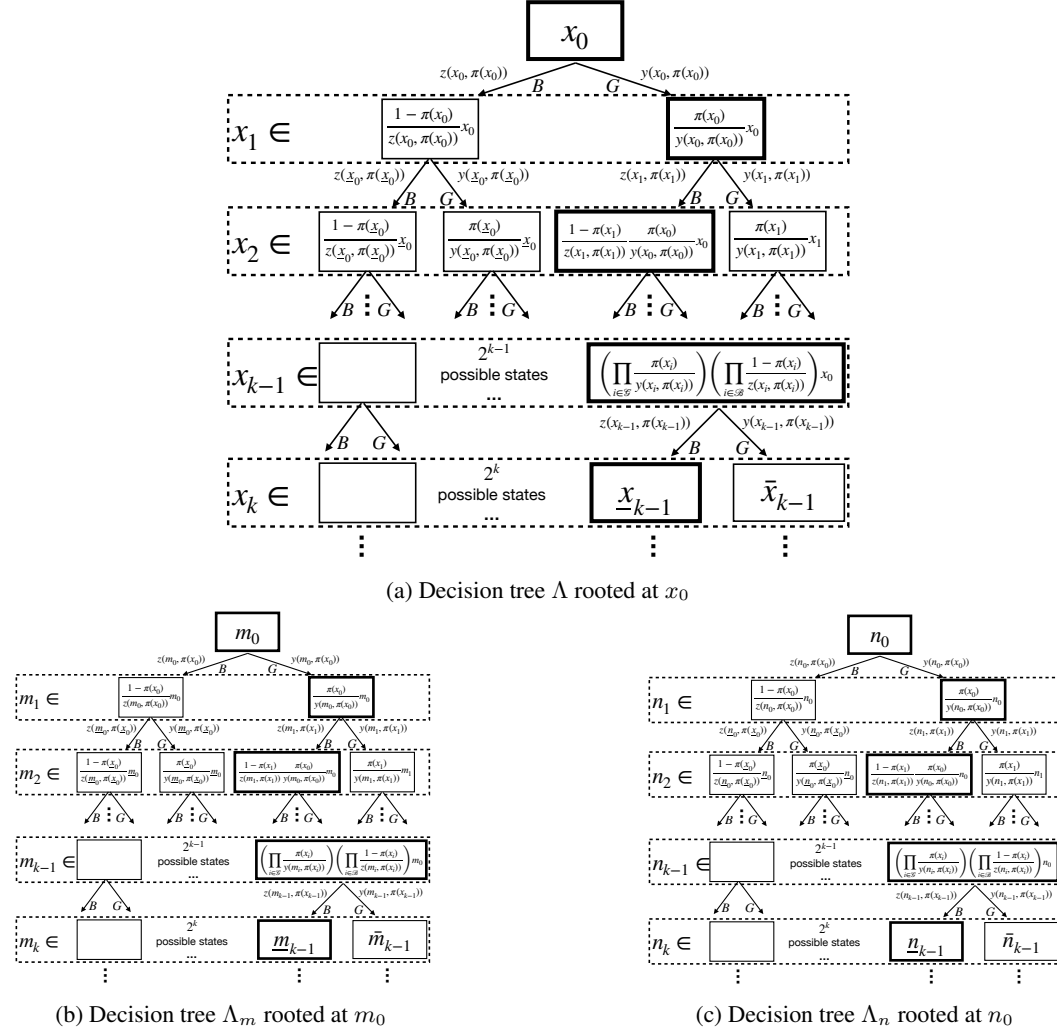


Figure 3: Here we depict the evolution of the belief in the form of three binary trees. Each tree originates, respectively, at the state x_0 , m_0 , and n_0 , and level i comprises the 2^i possible states reachable after i epochs. Left branches correspond to action B , yielding the updated belief $\underline{x}_0$ (respectively, $\underline{m}_0$ and $\underline{n}_0$), whereas right branches correspond to action G , yielding the updated belief $\bar{x}_0$ (respectively, $\bar{m}_0$ and $\bar{n}_0$). Each branch is labeled with the corresponding probability, and the bold states delineate an example path, such as λ . Along the example path, $0 \in \mathcal{G}$ and $1, k-1 \in \mathcal{B}$. The bold states at each level of the three trees are counterparts.

Now let $\phi_k^\pi(b_0) = \mathbb{E}[r_A(b_k, \pi(b_k))]$ denote the expected instantaneous reward at epoch k upon commencing with state b_0 and applying policy π , and set $\phi^*(x) = \phi^{\pi_A^*}(x)$. Invoking Fubini's Theorem, we may rewrite our

value function as

$$V_A^\pi(b_0) = \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r_A(b_i, \pi(b_i)) \right] = \sum_{i=0}^{\infty} \delta^i \mathbb{E}[r_A(b_i, \pi(b_i))] = \sum_{i=0}^{\infty} \delta^i \phi_i^\pi(b_0) \quad (25)$$

We may therefore rewrite (24) as

$$\sum_{i=0}^{\infty} \delta^i \phi_i^*(x_0) \leq t \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(m_0) \right] + (1-t) \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(n_0) \right] \quad (26)$$

To establish this, we show that, for every $i = 0, 1, 2, \dots$,

$$\phi_i^*(x_0) \leq t \phi_i^{\tilde{\pi}}(m_0) + (1-t) \phi_i^{\tilde{\pi}}(n_0) \quad (27)$$

which leads directly to (26), (24), and, ultimately, the convexity of $V^*(\cdot)$ with respect to the public belief.

Let us now consider a general $i = k$. The quantity $\phi_{k-1}^\pi(x_0)$ comprises 2^{k-1} terms, each corresponding to one path of possible actions within the set $\{G, B\}^{k-2}$. Thus, at epoch $k-1$, the path λ contributes the following term to $\phi_{k-1}^*(x_0)$:

$$\left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right) r_A(x_{k-1}, \pi_A^*(x_{k-1})) \quad (28)$$

We now define $P_\lambda^*(x_0) = \left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right)$ —that is, the probability of traversing the specified path λ upon commencing with state x_0 and following policy π_A^* . Once more, we shall use $\bar{x}_i$ and $\underline{x}_i$ to denote the beliefs updated from x_i following the observation of action G or B , respectively; thus $\bar{x}_i$ and $\underline{x}_i$ constitute the belief values of the children nodes (in the $(i+1)$ -th level) of the node with belief value x_i in the i -th level (see levels $k-1$ and k in figure 3a). Considering the children $\bar{x}_{k-1}$ and $\underline{x}_{k-1}$ (in the k -th level) of the node x_{k-1} (in the $(k-1)$ -th level), we obtain the following two terms in $\phi_k^*(x_0)$:

$$P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (29)$$

Analogously, we define $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \tilde{\pi}(m_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \tilde{\pi}(m_i)) \right)$ —the probability of traversing the specified path λ upon commencing with state m_0 and following policy $\tilde{\pi}$. Because $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, we have $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \pi_A^*(x_i)) \right)$. We define $P_\lambda^{\tilde{\pi}}(n_0)$ correspondingly, replacing $\{m_i\}$ with $\{n_i\}$.

The corresponding terms in $\phi_k^{\tilde{\pi}}(m_0)$ and $\phi_k^{\tilde{\pi}}(n_0)$ are

$$P_\lambda^{\tilde{\pi}}(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (30)$$

$$P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (31)$$

We shall show that

$$\begin{aligned} & P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & \leq t P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & \quad + (1-t) P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \end{aligned} \quad (32)$$

Analogously, each pair of children for each term in the $(k-1)$ -th level of the belief tree may be combined, and the corresponding inequalities obtained by the same procedure. Summing the resulting inequalities yields (27), which leads to (26), (24), and the convexity of $V^*(\cdot)$.

Proof of (32)

We now direct our attention to the latter of the two terms of (29). We shall establish the following two properties:

$$\begin{aligned} \underline{x}_{k-1} &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ & \quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \end{aligned} \quad (33)$$

$$\begin{aligned}
1 &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
&+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)
\end{aligned} \tag{34}$$

Defining $\tilde{t} = t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)$, it follows from (33) and (34) that

$$\underline{x}_{k-1} = \tilde{t} \underline{m}_{k-1} + (1-\tilde{t}) \underline{n}_{k-1} \tag{35}$$

From the expressions for $\tilde{t}$, $P_\lambda^*(x_0)$, $P_\lambda^{\tilde{\pi}}(m_0)$, it follows that:

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = t z(m_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(m_0) \tag{36}$$

Similarly, we can show

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = (1-t) z(n_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(n_0) \tag{37}$$

We apply this to the latter term of (29) as follows:

$$\begin{aligned}
&P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \\
&\leq P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) \\
&\quad [\tilde{t} r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) + (1-\tilde{t}) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Convexity of } r, \text{Eqn 35}) \\
&= t [P_\lambda^{\tilde{\pi}}(m_0) z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Eqns 36, 37})
\end{aligned} \tag{38}$$

Applying the identical argument to the first term of (29) yields:

$$\begin{aligned}
&P_\lambda^*(x_0) y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \\
&\leq t [P_\lambda^{\tilde{\pi}}(m_0) y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1}))]
\end{aligned} \tag{39}$$

Summing (38) and (39) yields (32).

We now bring the proof of (32)—and hence the entire proof—to completion by establishing (33) and (34).

Proof of (33)

From the recurrence relation in Equation (13) and Equation (3), we may express $\underline{x}_{k-1}$, $\underline{m}_{k-1}$, and $\underline{n}_{k-1}$ as the product of all belief updates with x_0 , m_0 , and n_0 , respectively, as follows, where each term in the products corresponds to the multiplicative state update effected by a single action.

$$\underline{x}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) x_0 \tag{40}$$

$$\underline{m}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(m_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(m_i, \pi_A^*(x_i))} \right) m_0 \tag{41}$$

$$\underline{n}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(n_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(n_i, \pi_A^*(x_i))} \right) n_0 \tag{42}$$

These equations are perhaps best apprehended via the illustration in figure 3. We then reorganize terms so as to relate $\underline{x}_{k-1}$ to $\underline{m}_{k-1}$ and $\underline{n}_{k-1}$, exploiting the fact that $x_0 = tm_0 + (1-t)n_0$.

$$\begin{aligned}\underline{x}_{k-1} &= \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) [tm_0 + (1-t)n_0] \quad (\text{from 40}) \\ &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \quad (\text{from 41 and 42}) \quad (43)\end{aligned}$$

This establishes (33).

Proof of (34)

We consider first the numerator of the first term in (34): $\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(m_i, \pi_A^*(x_i))$. We now prove that this may be written as $\alpha_0 m_0 + \beta_0$ for some α_0, β_0 that depend solely upon $\{\pi_A^*(x_i)\}_{i=1, \dots, k-1}$, the nature of the dependence being dictated by the sequence of actions along this path in stages 1 through $k-1$. We prove this by induction, working backward from the final term (i.e., the $(k-1)$ -th term) of the product; our base case is accordingly the final term of the product, which is either $y(m_{k-1}, \pi_A^*(x_{k-1}))$ or $z(m_{k-1}, \pi_A^*(x_{k-1}))$, depending upon the preceding action (in the $(k-1)$ -th stage).

$$\begin{aligned}y(m_{k-1}, \pi_A^*(x_{k-1})) &= 1 + 2m_{k-1}\pi_A^*(x_{k-1}) - m_{k-1} - \pi_A^*(x_{k-1}) \\ &= (2\pi_A^*(x_{k-1}) - 1)m_{k-1} + (1 - \pi_A^*(x_{k-1})) \\ z(m_{k-1}, \pi_A^*(x_{k-1})) &= m_{k-1} + \pi_A^*(x_{k-1}) - 2m_{k-1}\pi_A^*(x_{k-1}) \\ &= (1 - 2\pi_A^*(x_{k-1}))m_{k-1} + (\pi_A^*(x_{k-1}))\end{aligned}$$

The final term of the product is thus of the form $\alpha_{k-1}m_{k-1} + \beta_{k-1}$, where α_{k-1} and β_{k-1} depend solely upon $\pi_A^*(x_{k-1})$, and the nature of the dependence is dictated entirely by the $(k-1)$ -th action (since this action determines whether the final term is $y(\cdot)$ or $z(\cdot)$).

For the inductive step, suppose that $\prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i))$ may be written as $\alpha_{j+1}m_{j+1} + \beta_{j+1}$ for some $\alpha_{j+1}, \beta_{j+1}$ depending solely upon $\{\pi_A^*(x_i)\}_{i > j}$, the nature of the dependence being dictated entirely by the sequence of actions along this path from $i > j$ to $i = k-1$. We shall now show that $\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i))$ may be written as $\alpha_j m_j + \beta_j$, again with α_j, β_j depending solely upon $\{\pi_A^*(x_i)\}_{i > j-1}$, the nature of the dependence being dictated by the sequence

of actions along this path from $i > j-1$ to $i = k-1$. We define $f(b, q, a) = \begin{cases} y(b, q) & a = G \\ z(b, q) & a = B \end{cases}$ and

$$g(q, a) = \begin{cases} q & a = G \\ 1 - q & a = B \end{cases}. \text{ Thus,}$$

$$\begin{aligned}&\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) [\alpha_{j+1}m_{j+1} + \beta_{j+1}] \quad (\text{inductive hypothesis}) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \left[\alpha_{j+1} \frac{g(\pi_A^*(x_j), a_{j+1})}{f(m_j, \pi_A^*(x_j), a_{j+1})} m_j + \beta_{j+1} \right] \quad (\text{Equation (13)}) \\ &= \alpha_{j+1}g(\pi_A^*(x_j), a_{j+1})m_j + \beta_{j+1}f(m_j, \pi_A^*(x_j), a_{j+1})\end{aligned}$$

Thus, from the definitions of $f(\cdot)$, $g(\cdot)$, $y(\cdot)$, and $z(\cdot)$,

$$\alpha_j = \begin{cases} \alpha_{j+1}\pi_A^*(x_j) + \beta_{j+1}(2\pi_A^*(x_j) - 1) & a_{j+1} = G \\ \alpha_{j+1}(1 - \pi_A^*(x_j)) + \beta_{j+1}(1 - 2\pi_A^*(x_j)) & a_{j+1} = B \end{cases} \text{ and } \beta_j = \begin{cases} \beta_{j+1}(1 - \pi_A^*(x_j)) & a_{j+1} = G \\ \beta_{j+1}\pi_A^*(x_j) & a_{j+1} = B \end{cases}$$

In either case, by our inductive hypothesis, $\alpha_{j+1}, \beta_{j+1}$ depend solely upon $\{\pi_A^*(x_i)\}_{i > j}$ and the actions along this path from $i > j$ to $i = k-1$. The claim thus holds for α_j, β_j , which completes the induction.

Now recall that the denominator of (34) is $\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(x_i, \pi_A^*(x_i))$. Observe that the policy and the sequence of actions coincide with those for the numerator; hence, because the coefficients α_0, β_0

depend, as shown, solely upon $\{\pi_A^*(x_i)\}$ and the path of actions, the same reasoning permits us to write the denominator as $\alpha_0 x_0 + \beta_0$ for the same α_0, β_0 . An identical argument applies to the numerator of the second term in (34). We may thus reason about the coefficients in (33) as follows:

$$\begin{aligned}
& t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& = \frac{t(\alpha_0 m_0 + \beta_0) + (1-t)(\alpha_0 n_0 + \beta_0)}{\alpha_0 x_0 + \beta_0} \\
& = \frac{\alpha_0 x_0 + \beta_0}{\alpha_0 x_0 + \beta_0} \quad (\text{Since } x_0 = t m_0 + (1-t) n_0) \\
& = 1
\end{aligned}$$

Thus (34) holds, which completes the proof. $\square$

C.4 ALTRUISTIC MYOPIC POLICY BOUND

Lemma 9.

$$\pi_A^0(b) \leq \pi_A^*(b) \quad \forall b \in [0, 1]$$

Suppose, by way of contradiction, that there exists $b \in [0, 1]$ such that $\pi_A^0(b) > \pi_A^*(b)$. We shall demonstrate that $V_A^*(b) < V_A^\pi(b)$ for a policy π that we construct, thereby contravening the optimality of $V_A^*(\cdot)$.

Let π be the policy that, commencing at public belief b , applies precision $\pi_A^0(b)$ at the current time step and thereafter applies the optimal policy at every future time step. We now expand $V_A^*(b) - V_A^\pi(b)$ by decomposing the value function into the instantaneous reward r_A and the future discounted reward. We shall denote by b^+ and b^- the beliefs updated upon receiving signals G and B , respectively, under the optimal policy, and by b_m^+ and b_m^- the analogous quantities under the myopic policy.

$$\begin{aligned}
V_A^*(b) - V_A^\pi(b) &= r_A(b, \pi_A^*(b)) - r_A(b, \pi_A^0(b)) \\
&+ \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \\
&< \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \quad (\text{definition of } \pi_A^0) \\
&\leq 0 \quad (\text{Theorem 2})
\end{aligned}$$

Here we invoked the fact that, by the definition of the myopic optimal policy, the instantaneous reward under $\pi_A^0(\cdot)$ must be the greater.

We then relied, for the future cost, upon the convexity of $V_A^*(\cdot)$ established in Theorem 2. From Equation (13), note that $b_m^- \leq b^- \leq b \leq b^+ \leq b_m^+$; this, in conjunction with the aforementioned convexity, yields the final step above and completes the proof. $\blacksquare$

C.5 PROOF OF THEOREM 3: OPTIMAL ALTRUISTIC POLICY

Consider $V_A^\pi(b)$ under the policy π that applies precision q at the current time step and the optimal policy at every future time step. The value of q that maximizes $V_A^\pi(b)$ furnishes the optimal precision commencing at public belief b . Since the planner never derives benefit from decreasing precision below the baseline p (see Appendix C.9), we restrict q to $[p, 1]$ without loss of generality.

We shall let b^+ and b^- denote the positive and negative belief updates that may arise upon receiving a signal of precision q . Taking the second derivative of $V_A^\pi(b)$ with respect to q for $q \geq \max(b, 1-b)$:

$$\frac{\partial^2}{\partial q^2} V_A^\pi(b) = \begin{cases} -\ddot{\beta}(q) + \delta b^2 (1-b)^2 \left[\frac{\ddot{V}_A^*(b^+)}{y^q(b, q)} + \frac{\ddot{V}_A^*(b^-)}{z^q(b, q)} \right] & q \geq \max(b, 1-b) \\ -\ddot{\beta}(q) & \text{otherwise} \end{cases}$$

By virtue of the convexity of $V_A^*(\cdot)$ and the concavity of $\beta(\cdot)$, respectively, the foregoing establishes that $V_A^\pi(b)$ is convex with respect to q on $[\max(b, 1-b), 1]$; on this interval it is therefore maximized at one of the two

extreme points. For $q \in [0.5, \max(b, 1 - b))$, the derivative of $V_A^\pi(b)$ with respect to q is negative, so that the optimal choice is the baseline precision p , which minimizes the cost. There are thus three candidate optimal precisions: p , $\max(b, 1 - b)$, and 1.

Observe that the expected value under the latter two candidates is invariably strictly negative, whereas under precision p the expected value equals 0 at $b = 0$ and $b = 1$. This, together with the non-positivity and symmetry of the value function (Lemma 8), entails the existence of a d_A such that $\pi_A^*(b) = p$ for $b \in [0, d_A) \cup (1 - d_A, 1]$. Moreover, Theorem 1 in conjunction with Lemma 9 yields $d_A \in (0, t_M)$, where t_M is as defined in the statement of Theorem 1.

Finally, Theorem 1 together with Lemma 9 likewise entails the existence of a $t_A \in [d_A, t_M]$ such that $\pi_A^*(b) = 1$ for $b \in (t_A, 1 - t_A)$. This completes the proof. $\blacksquare$

C.6 PROOF OF THEOREM 4: MYOPIC BIASED POLICY

Preliminaries Applying the action rule of (2), the instantaneous reward (6) may be written as follows:

$$r_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) - C & b < 1 - q \\ -\beta(|q - p|) & b > q \end{cases} \quad (44)$$

Note that $z(b, q) = b + q - 2bq$ increases with respect to q when $b < 0.5$ and decreases with respect to q when $b > 0.5$.

At $b = 0.5$, we have $z(b, q) = z(0.5, q) = 0.5$ irrespective of the precision chosen. Moreover, at this public belief, $q \geq \max(b, 1 - b)$ (since $q \geq 0.5$ for any q), placing us in the first case of (44). In this case the second term of $r_B(b, q)$ is invariant in q , so that the planner seeks to maximize the first term alone, which is achieved at $q = p$.

We now establish a series of lemmas (10–14) that will subserve our proof of Theorem 4.

Lemma 10.

If $\pi_B^0(b) \geq \max(b, 1 - b)$, then $\pi_B^0(b) \in \{\max(b, 1 - b), p, 1\}$.

Proof. The reward is differentiable everywhere save at $q = p$. We may therefore write its derivative with respect to the chosen precision as follows (except at $q = p$):

$$\frac{\partial r_B(b, q)}{\partial q} = \begin{cases} -\text{sign}(q - p)\dot{\beta}(|q - p|) & q \geq \max(b, 1 - b) \\ -C(1 - 2b) & \\ -\text{sign}(q - p)\dot{\beta}(|q - p|) & \text{otherwise} \end{cases} \quad (45)$$

In the former case, we may write the second derivative of the reward with respect to q as follows, for $q \neq p$:

$$\frac{\partial^2}{\partial q^2} r_B(b, q) = -\ddot{\beta}(|q - p|)$$

Since $\beta(\cdot)$ is concave, this expression is non-negative, whence $r_B(b, q)$ is convex with respect to q . The reward is thus maximized at one of the extreme points $\{\max(b, 1 - b), 1\}$ or at p , the latter because $r_B(b, q)$ fails to be differentiable at $q = p$. $\square$

Lemma 11.

For $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. Since $p > 0.5$, for $b \in [0, 1 - p)$ we have $\max(b, 1 - b) = 1 - b > p$.

Consider first the case in which $\pi_B^0(b) < \max(b, 1 - b) = 1 - b$. From (44), in such instances the reward is maximized by minimizing $|q - p|$; the optimal precision below $\max(b, 1 - b)$ is therefore p .

Consider now $\pi_B^0(b) \geq \max(b, 1 - b) = 1 - b$. By Lemma 10, $\pi_B^0(b) \in \{1 - b, p, 1\}$, and it remains to exclude $\pi_B^0(b) = 1$. To this end, we compare $r_B(b, 1)$ and $r_B(b, 1 - b)$, both of which fall under the first case of (44). Since $\beta(\cdot)$ is increasing in its argument and $p < 1 - b \leq 1$, the first term of the reward in (44) is greater for precision $1 - b$ than for precision 1. Since, moreover, $z(b, q)$ is increasing in q for $b < 0.5$, we have $r_B(b, 1) \leq r_B(b, \max(b, 1 - b))$, and precision 1 may thus be excluded. This completes the proof. $\square$

Lemma 12.

For $b \in (p, 1]$, $\pi_B^0(b) = p$.

Proof. On account of the term $-\beta(|q - p|)$, the reward is strictly negative for any $q \neq p$. For $b \in (p, 1]$ and $q = p$, we have $p < \max(b, 1 - b) = b$, which places us in the third case of (44) and yields reward 0. The optimal precision is therefore p . $\square$

Lemma 13.

For $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. When $p = 0.5$, the interval $[1 - p, 0.5]$ reduces to the singleton $\{0.5\}$; at $b = 0.5$ we have $p = 1 - b$, whence $\pi_B^0(b) = p = 1 - b = 0.5$. For the remainder of the proof we treat $p > 0.5 \geq b$.

For $b \in [1 - p, 0.5]$, we have $\max(b, 1 - b) = 1 - b \leq p$; hence, from the first case of (44), precision p yields reward $r_B(b, p) = -Cz(b, p)$. We now distinguish two cases: $q < \max(b, 1 - b)$ and $q \geq \max(b, 1 - b)$.

Suppose $q < \max(b, 1 - b) = 1 - b$. This places us in the second case of (44) and yields reward $-\beta(|q - p|) - C$. Moreover, $z(b, q)$ is increasing in q for $b < 0.5$ and assumes values between $z(b, 0.5) = 0.5$ and $z(b, 1) = 1 - b$, whence $z(b, q) < 1$ for all q . The reward from a policy $q < 1 - b$ is therefore strictly lower than $r_B(b, p)$, and this case may be excluded.

When $q \geq \max(b, 1 - b)$, an application of Lemma 10 yields the remaining candidate precisions $\{1 - b, p, 1\}$. We exclude precision 1 by comparison with precision p : the first term of the reward in (44) is greater for precision p than for precision 1, and, since $z(b, q)$ is increasing in q for $b < 0.5$, the second term is likewise greater for precision p . Precision 1 may thus be excluded, which completes the proof. $\square$

We now introduce the notion of ϵ -optimal policies.

Definition 1. ϵ -optimal policies

An ϵ -optimal policy $\pi^\epsilon(b)$ is any policy such that the following holds

$$V^*(b) \leq V^{\pi^\epsilon}(b) + \epsilon$$

for $\epsilon > 0$.

Note that, by the definition, any optimal policy is likewise ϵ -optimal for every $\epsilon > 0$; in particular, $\pi_B^0(b)$ is ϵ -optimal as well.

In accordance with this definition, let $\pi_B^{\epsilon,0}(b)$ denote an ϵ -optimal myopic policy for the biased planner.

Lemma 14.

For $b \in (0.5, p]$ and any $\epsilon > 0$, there are two possibilities:

1. $\pi_B^0(b) \in \{p, 1\}$, or
2. $\pi_B^{\epsilon,0}(b) = b - \epsilon$

Proof. For $b \in (0.5, p]$, we have $\max(b, 1 - b) = b \leq p$. We distinguish two cases: $q \geq \max(b, 1 - b)$ and $q < \max(b, 1 - b)$.

In the first case, Lemma 10 yields $\pi_B^0(b) \in \{b, p, 1\}$. We exclude precision b by comparison with precision p : the first term of the reward in (44) is greater for precision p , and, since $z(b, q)$ is decreasing in q for $b > 0.5$, the second term is likewise greater for precision p . Precision b may thus be excluded, which yields the first claim.

In the latter case, $q < \max(b, 1 - b) = b$. Consequently, (1) since $q \geq 0.5$, we have $q \in [0.5, b)$, and (2) the third case of (44) applies, yielding reward $-\beta(|q - p|)$. The reward $r_B(b, q)$ is thus an increasing function of $q \in [0.5, b)$, since $0.5 < b \leq p$. This yields directly the second claim. $\square$

Remark 3. As noted in Puterman (1990), Section 2.3.1, an optimal policy may fail to exist for infinite action spaces, inasmuch as the supremum may not be attained. When no optimal policy exists, we instead seek ϵ -optimal policies—that is, policies yielding reward within $\epsilon > 0$ of the supremum. This is the case, in particular, for case 2 of Lemma 14.

Proof of Theorem 4 We treat the myopic biased policy across four cases— $b \in [0, 1-p]$, $b \in [1-p, 0.5]$, $b \in (0.5, p]$, and $b \in (p, 1]$ —availing ourselves of the fact that $p \geq 0.5$. In light of Lemmas 11–14, these cases admit, respectively, the following candidate policies: $\{p, 1-b\}$, $\{1-b, p\}$, $\{1, p, b-\epsilon\}$, and $\{p\}$.

Case 1: $b \in [0, 1-p]$, $\pi_B^0(b) \in \{p, 1-b\}$ (Lemma 11)

We shall establish the existence of a threshold t_1 such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1-b$ for $b \in (t_1, 1-p]$.

Applying (44), we may express the reward under the two candidate policies as follows:

$$\begin{aligned} r_B(b, p) &= -C \\ r_B(b, 1-b) &= -\beta(1-b-p) - C(1-2b(1-b)) \end{aligned}$$

The derivative of $r_B(b, 1-b)$ with respect to b is

$$\frac{\partial}{\partial b} r_B(b, 1-b) = \dot{\beta}(1-p-b) + 2C(1-2b)$$

which is positive, since $\beta(\cdot)$ is increasing and $b < 1-p \leq 0.5$. Moreover, upon evaluation at $b = 0$ and $b = 1-p$, we obtain

$$\begin{aligned} r_B(0, 1) &= -\beta(1-p) - C < r_B(b, p) \\ r_B(1-p, p) &= -C(1-2p(1-p)) > r_B(b, p) \end{aligned}$$

Thus $r_B(b, 1-b)$ is increasing on $[0, 1-p]$, with $r_B(b, 1-b) < r_B(b, p)$ at $b = 0$ and the reverse at $b = 1-p$. Since $\beta(\cdot)$ is continuous, so too are both reward functions; there consequently exists a $t_1 \in (0, 1-p)$ such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1-b$ for $b \in (t_1, 1-p]$.

Case 2: $b \in [1-p, 0.5]$, $\pi_B^0(b) \in \{1-b, p\}$ (Lemma 13)

If $p = 0.5$, this interval reduces to the singleton $\{0.5\}$, and, at $b = 0.5$, we have $1-b = p$; the two candidate policies thus coincide, and the optimal policy is $\pi_B^0(b) = p$. For the remainder of this case, we consider $p > 0.5$.

We now establish the existence of t_2, t_3 such that $\pi_B^0(b) = 1-b$ for $b \in [t_2, t_3]$ and $\pi_B^0(b) = p$ for $b \in [1-p, t_2] \cup [t_3, 0.5]$.

Applying (44), we may express the reward under the candidate policies as follows:

$$\begin{aligned} r_B(b, 1-b) &= -\beta(p-1+b) - Cz(b, 1-b) \\ r_B(b, p) &= -Cz(b, p) \end{aligned}$$

This gives rise to the following condition for the optimal policy to be $1-b$:

$$\begin{aligned} r_B(b, 1-b) &\geq r_B(b, p) \\ \beta(p-1+b) &\leq C[z(b, p) - z(b, 1-b)] \\ \beta(p-1+b) &\leq C(p-1+b)(1-2b) \end{aligned} \tag{46}$$

The LHS of (46) is a concave, increasing function of b on $[1-p, 0.5]$, whereas the RHS of (46) is a concave parabola that increases over the first half of $[1-p, 0.5]$ and decreases over the second half.

Consider now the two halves of the interval, $[1-p, 0.75-0.5p]$ and $[0.75-0.5p, 0.5]$. Over the first half, both the LHS and the RHS of (46) are increasing and concave, the RHS being additionally a parabola; the two therefore intersect at most twice (one intersection occurring at $b = 1-p$). Over the second half, the LHS of (46) is increasing while the RHS is decreasing, so that they intersect at most once. Let the two possible intersections for $b > 1-p$ be denoted t_2 and t_3 , with $t_2 \leq t_3$. It remains to determine the optimum on each of $(1-p, t_2)$, (t_2, t_3) , and $(t_3, 0.5)$. At $b = 0.5$, we have $r_B(b, p) > r_B(b, 1-b)$ (since $p > 0.5$), whence $\pi_B^0(b) = p$ for $b \in (t_3, 0.5)$. It follows that $\pi_B^0(b) = 1-b$ on (t_2, t_3) and $\pi_B^0(b) = p$ on $(1-p, t_2)$.

Case 3: $b \in (0.5, p]$, $\pi_B^{\epsilon, 0}(b) \in \{1, p, b-\epsilon\}$ (Lemma 14)

In this case, we establish the existence of $0.5 \leq t_4 \leq t_5 \leq p$ such that the optimal precisions on the intervals $(0.5, t_4]$, (t_4, t_5) , and $[t_5, p)$ are p , 1 , and $b-\epsilon$, respectively.

Applying (44), we may express the reward under the candidate policies as follows:

$$\begin{aligned} r_B(b, 1) &= -\beta(1-p) - C(1-b) \\ r_B(b, p) &= -Cz(b, p) \\ r_B(b, b-\epsilon) &= -\beta(p-b+\epsilon) \end{aligned}$$

When evaluated at $b = p$, $r_B(b, b - \epsilon)$ is arbitrarily close to 0; hence $b - \epsilon$ is the optimal precision on some interval $(t_5, p]$ for $0.5 \leq t_5 < p$.

Now observe that $r_B(b, 1)$ and $r_B(b, p)$ are linear functions of b , whereas $r_B(b, b - \epsilon)$ is convex in b (by virtue of the concavity of β). Consequently, $r_B(b, b - \epsilon) - r_B(b, p)$ and $r_B(b, b - \epsilon) - r_B(b, 1)$ are convex and thus possess convex level sets. This implies that the set on which both expressions are positive—that is, the set on which $b - \epsilon$ is the optimal policy—is likewise convex. Accordingly, $b - \epsilon$ cannot be optimal on any subset of $(0.5, t_5)$.

It remains to compare precisions 1 and p on the interval $(0.5, t_5]$. To this end, we inspect the values of $r_B(b, 1)$ and $r_B(b, p)$ at 0.5 together with their derivatives.

$$\begin{aligned} r_B(0.5, 1) &= -\beta(1 - p) - 0.5C \\ r_B(0.5, p) &= -0.5C \end{aligned}$$

Since $r_B(0.5, 1) < r_B(0.5, p)$, there exists an interval $(0.5, t_4)$ with $t_4 \leq t_5$ on which $\pi_B^0(b) = p$. Moreover, since both functions are linear in b , there is at most one crossover point at which their ordering reverses; hence, if $t_4 < t_5$, then $\pi_B^0(b) = 1$ on (t_4, t_5) .

Case 4: $b \in (p, 1]$, $\pi_B^0(b) = p$

This follows immediately from Lemma 12 and completes the proof of Theorem 4. ■

C.7 BIASED VALUE FUNCTION MONOTONICITY

Lemma 15. *For any $b_1, b_2 \in [0, 1]$ such that $b_1 \leq b_2$,*

$$V_B^*(b_1) \leq V_B^*(b_2)$$

Proof. Applying (44), the derivative of the biased reward with respect to b may be written as follows:

$$\frac{\partial}{\partial b} r_B(b, q) = \begin{cases} -C(1 - 2q) & q \geq \max(b, 1 - b) \\ 0 & \text{otherwise} \end{cases} \quad (47)$$

Since $q \geq 0.5$ and $z(b, q) \in [0, 1]$, it follows that $r_B(b, q)$ is increasing with respect to b .

Next, observe that the Bayesian character of the belief updates entails that the belief process is a martingale—that is, for any fixed q_i , $\mathbb{E}[b_{i+1}] = b_i$. To demonstrate this, consider two cases: (1) $q_i < \max(b_i, 1 - b_i)$, and (2) $q_i \geq \max(b_i, 1 - b_i)$.

In the former case, (3) yields $b_{i+1} = b_i$, which satisfies the claim.

In the latter case, (13) supplies the distribution of b_{i+1} , which yields the same conclusion.

The monotonicity of $V_B^*(\cdot)$ then follows from Proposition 5 of Smith & McCardle (2002). □

C.8 PROOF OF THEOREM 5: OPTIMAL BIASED POLICY

We first establish, in the following lemma, that the optimal value function is continuous:

Lemma 16. *Biased value function continuity*

$$V_B^*(\cdot) \text{ is continuous w.r.t. } b.$$

Proof. Consider first the one-stage value function, defined as $V_B^0(b) = \sup_{q \in [0.5, 1]} r_B(b, q)$. From Theorem 4, we know that $\pi_B^{\epsilon, 0}(b) \in \{1 - b, 1, b - \epsilon, p\}$; we may therefore write the one-stage value function equivalently as follows:

$$\begin{aligned} V_B^0(b) &= \sup_{\epsilon > 0} r_B(b, \pi_B^{\epsilon, 0}(b)) \\ &= \sup_{\epsilon > 0} \max_{q \in \{p, 1 - b, 1, b - \epsilon, p\}} r_B(b, q) \end{aligned} \quad (48)$$

By means of the foregoing expression, we now argue that $V_B^0(\cdot)$ is continuous on $b \in [0, 1]$. Consider first $b \leq 0.5$. By Theorem 4, an ϵ -optimal policy applies precision $1 - b$ or p for $b \leq 0.5$; in either case the instantaneous reward is independent of ϵ , so that the supremum in (48) may be dispensed with, leaving $V_B^0(b) = \max_{q \in \{1 - b, p\}} r_B(b, q)$. For $b \leq 0.5$, $r_B(b, 1 - b)$ falls under case one of (44) and is thus continuous in b . On the same interval, $r_B(b, p)$ exhibits a single discontinuity at $b = 1 - p$, where it transitions from case two to

case one of (44). At $b = 1 - p$, Theorem 4 shows that the optimal myopic policy switches from $1 - b$ to p ; both policies fall under the first case of (44), and, at $b = 1 - p$, we have $p = 1 - b$, so that the controls coincide. The left and right limits thus agree with the optimal value, and $V_B^0(\cdot)$ is continuous.

For $b \geq 0.5$, Theorem 4 furnishes the candidate ϵ -optimal policies $q \in \{1, b - \epsilon, p\}$. When the inner maximum in (48) is attained at $q = b - \epsilon$, the resulting value is $V_B^0(b) = \sup_{\epsilon > 0} r_B(b, b - \epsilon)$; by the third case of (44), we may rewrite this as $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|b - \epsilon - p|)$. Since $b - \epsilon$ is selected only for $b \leq p$ (Theorem 4), this reduces to $V_B^0(b) = -\beta(|p - b|)$. Hence, interchanging the maximum and the supremum in (48), we obtain $V_B^0(b) = \max\{r_B(b, 1), -\beta(|p - b|), r_B(b, p)\}$ for $b \geq 0.5$. As to the first expression in this maximum, $r_B(b, 1)$ falls under case one of (44) and is thus continuous; the continuity of $-\beta(|p - b|)$ follows from that of $\beta(\cdot)$. As to $r_B(b, p)$, since $b \geq 0.5$, the first case of (44) applies for $b \leq p$ and the third for $b > p$, so that $r_B(b, p)$ is potentially discontinuous at $b = p$. It remains, therefore, to establish the continuity of $V_B^0(\cdot)$ at $b = p$.

At $b = p$, the myopic ϵ -optimal policy switches from $q = b - \epsilon$ to $q = p$ by Theorem 4, with $\pi_B^{\epsilon, 0}(p) = b - \epsilon$. Thus, for $b = p$, the inner maximum of (48) is attained at $q = b - \epsilon$ and, by the third case of (44), $V_B^0(p) = \sup_{\epsilon > 0} -\beta(\epsilon) = 0$. Similarly, by Theorem 4, for each $\epsilon > 0$ there exists $t_5 < p$ such that on $[t_5, p]$ the inner maximum of (48) is attained by $q = b - \epsilon$; on this interval, $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|p - b + \epsilon|) = -\beta(|p - b|)$, whence $\lim_{b \uparrow p} V_B^0(b) = 0$. For $b > p$, the myopic optimal policy is $q = p$ by Theorem 4, which yields reward $V_B^0(b) = 0$ for all $b > p$ by the third case of (44). This establishes the continuity of $V_B^0(\cdot)$ at $b = p$ and completes the proof that $V_B^0(\cdot)$ is continuous on $[0, 1]$.

An application of Theorem 4.2 of Hinderer (2005) then yields the continuity of $V_B^*(\cdot)$. $\square$

Remark 4. Given the continuity of $V_B^*(\cdot)$ with respect to the public belief, together with the fact that the control variable resides in the compact set $[0.5, 1]$, one might be tempted to conclude that an optimum is guaranteed to exist at every public belief. There is, however, an important subtlety that forecloses such a conclusion: although $V_B^*(\cdot)$ is continuous with respect to the public belief, it need not be continuous with respect to the precision. Specifically, b_{i+1} is a function of both b_i and q_i (see Equation (3)), so that $\mathbb{E}[V_B^*(b_{i+1})]$ may be denoted $\phi(b_i, q_i)$; now, b_{i+1} need not be continuous in q_i , whence $\phi(\cdot)$ may fail to be continuous in q_i . There may accordingly exist regimes in which the optimal value is attained by no policy—and we shall indeed demonstrate that such cases arise.

Proof of Theorem 5 We now marshal Lemma 16 in the service of proving Theorem 5. We again let $z(b, q) = b + q - 2bq$ denote the probability of an agent receiving signal B under precision q and prior public belief b , and $y(b, q) = 1 - z(b, q)$ the corresponding probability of the agent receiving signal G . We abuse notation so as to let $V_B(b, q)$ denote the expected utility accruing when one commences at public belief b , applies precision q at the current time step, and thereafter follows the optimal policy $\pi_B^*(\cdot)$ at all future steps. Applying (13), (3), and (44), we may express $V_B(b, q)$ as follows:

$$V_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) \\ \quad + \delta y(b, q) V_B^*(f(b, q, G)) \\ \quad + \delta z(b, q) V_B^*(f(b, q, B)) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) \\ \quad - C\mathbf{1}(b < 0.5) & \text{otherwise} \\ + \delta V_B^*(b) \end{cases} \quad (49)$$

Consider the latter case of (49), in which $q < \max(b, 1 - b)$. Here the sole term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by choosing q as near to p as possible. Consequently, if the optimal precision falls below $\max(b, 1 - b)$, it will be as close to p as possible.

We now distinguish four cases: (1) $b \in [0, 1 - p]$, (2) $b \in [1 - p, 0.5]$, (3) $b \in [0.5, p]$, and (4) $b \in (p, 1]$.

Case 1: $b \in [0, 1 - p]$, claims (A) and (B)

In this case, we shall prove the existence of t_1 such that $\pi_B^*(b) = p$ for $b \leq t_1$. Note first that if $\pi^*(b) < \max(b, 1 - b)$, we are situated in the second case of Equation (49), wherein the sole term affected by the precision policy is $-\beta(|q - p|)$, maximized by minimizing $|q - p|$. Since $p \in [0.5, 1]$, we have $p < \max(b, 1 - b)$ whenever $b \in [0, 1 - p]$; hence, for $b \in [0, 1 - p]$, if $\pi_B^*(b) < \max(b, 1 - b)$, then $\pi_B^*(b) = p$.

We now evaluate the value function at the belief $b = 0$. At $b = 0$ the state of the world is known to be B , so that no signal, whatever its precision, will displace the public belief—a fact seen formally upon substituting $b = 0$ into the updated-belief expressions in (13). Furthermore, at $b = 0$, if $q \geq \max(b, 1 - b)$ then $q = 1$; and, as argued above, if the precision falls below $\max(b, 1 - b)$, the optimal precision is $q = p$. We may compare

the resulting value under these two candidate policies as follows, via Equation (49):

$$V_B(0, 1) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (50)$$

$$V_B(0, p) = -C + \delta V_B^*(0) \quad (51)$$

Since $V_B(0, 1) < V_B(0, p)$, we have $\pi_B^*(0) = p$ and $V_B^*(0) = V_B(0, p)$; hence, from Equation (51), $V_B^*(0) = \frac{-C}{1-\delta}$. By Lemma 15, this constitutes a lower bound for the value function at any public belief. Consequently, and since $V_B^*(b) \leq 0$, we have $|V_B^*(b)| \leq \frac{C}{1-\delta}$ for all $b \in [0, 1]$.

We now show that there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) falls short of that of the second case of Equation (49). On this interval, therefore, the optimal control resides in the second case of Equation (49), so that $\pi_B^*(b) < \max(b, 1 - b)$; and it follows from our initial argument earlier in this case that $\pi_B^*(b) = p$ throughout the interval, which would complete the proof.

We define the function $g(\cdot)$ to be the maximum of the first case of the RHS of Equation (49):

$$g(b) = \max_{q \geq \max(b, 1-b)} \left[-\beta(|q - p|) - Cz(b, q) + \delta y(b, q) V_B^*(f(b, q, G)) + \delta z(b, q) V_B^*(f(b, q, B)) \right] \quad (52)$$

For $b \in (0, 1)$, both $f(b, q, G)$ and $f(b, q, B)$ are continuous in q by Equation (13). The maximum in Equation (52) must therefore exist, being that of a function continuous in q over the compact set $q \in [\max(b, 1 - b), 1]$.

We now take the limit of $g(\epsilon)$ as ϵ tends to 0. Since $q \in [\max(\epsilon, 1 - \epsilon), 1]$, as $\epsilon \rightarrow 0$ we have $q \rightarrow 1$ (within the maximum in Equation (52)). We reason about each of the four terms in Equation (52) in turn. As $\epsilon \rightarrow 0$, $q \rightarrow 1$, so that: (1) by the continuity of $\beta(\cdot)$, $\beta(|q - p|) \rightarrow \beta(1 - p)$; and (2) by the continuity of $z(\cdot)$, $z(\epsilon, q) \rightarrow 1$ and $y(\epsilon, q) \rightarrow 0$, whence the second term tends to C . Since $V_B^*(\cdot)$ is bounded and $y(\epsilon, q) \rightarrow 0$, the third term tends to 0. Finally, since $z(\epsilon, q) \rightarrow 1$ and $f(\epsilon, q, B) \rightarrow 0$, and since $V_B^*(\cdot)$ is continuous by Lemma 16, the last term approaches $\delta V_B^*(0)$. Thus,

$$\lim_{\epsilon \rightarrow 0} g(\epsilon) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (53)$$

Applying precision p , however, yields the following from the second case of Equation (49):

$$\lim_{\epsilon \rightarrow 0} V_B(\epsilon, p) = \lim_{\epsilon \rightarrow 0} [-\beta(p - p) - C + \delta V_B^*(\epsilon)] \quad (54)$$

$$= -C + \delta V_B^*(0) \quad (55)$$

The final equality follows from the continuity of $V_B^*(\cdot)$ established in Lemma 16.

Observe that the expression in Equation (53) is strictly less than that in Equation (55). There consequently exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) falls short of that of the second case of Equation (49). The result follows.

Case 2: $b \in [1 - p, 0.5)$, Claim (C)

We now argue that for $b_i \in [1 - p, 0.5)$ we have $\pi^*(b_i) \geq 1 - b_i$, thereby yielding claim (C). Consider a stationary policy $\pi(\cdot)$ with $\pi(b_i) < 1 - b_i$. Note that, by Equation (3), the public belief remains unaltered under such a policy; the same control is therefore applied at every future time step, and, by Equation (2), every agent selects action B . Exploiting the facts that $1 - b_i \leq p$ throughout this belief range, that $\mathbb{P}(a_i = B | b_i, \pi(b_i)) = 1$, and that $b_{i+1} = b_i$, we may apply Equation (6) to write the expected total discounted utility under such a policy as follows:

$$V_B^\pi(b_i) = -\beta(p - \pi(b_i)) - C + \delta V_B^\pi(b_i)$$

Thus, if this policy is applied, it yields expected value $V_B^\pi(b_i) = \frac{-\beta(p - \pi(b_i)) - C}{1 - \delta}$ —that is, the cost incurred by every agent's choosing action B while precision $\pi(b_i)$ continues to be applied. This falls strictly below the value-function lower bound of $\frac{-C}{1 - \delta}$ established via Lemma 15 in Case 1, since $\pi(b_i) < 1 - b_i \leq p$. Hence, if an optimal policy exists at this point, it must satisfy $\pi_B^*(b_i) \geq 1 - b_i$.

The remaining candidate policies are $q_i \in [1 - b_i, 1]$. Invoking Bellman's principle, we may write the following:

$$V_B^*(b_i) = \sup_{q_i \in [1 - b_i, 1]} r_B(b_i, q_i) + \delta \mathbb{E}[V_B^*(b_{i+1})] \quad (56)$$

Since $q_i \geq \max(b_i, 1 - b_i)$ and $b \in [1 - p, 0.5) \subset (0, 1)$, Equation (13) and Equation (3) imply that $b_{i+1} = \frac{b_i q_i}{y(b_i, q_i)}$ or $b_{i+1} = \frac{b_i(1 - q_i)}{z(b_i, q_i)}$. Both of these expressions are continuous in q_i , and $V_B^*(\cdot)$ is continuous

by Lemma 16; the RHS of Equation (56) within the supremum is therefore continuous in q_i . Consequently, in Equation (56), the supremum over the compact set $q_i \in [1 - b, 1]$ of a function continuous in q_i must be attained, which implies the existence of an optimal policy. Thus $\pi_B^*(b_i) \geq 1 - b_i$.

Case 3: $b \in [0.5, p]$ Claims (D) and (E)

For $b = 0.5$, we have $q \geq b$ (since we assumed $q \in [0.5, 1]$), which places us in the first case of Equation (49). Moreover, $\beta(\cdot)$, $y(b, q)$, $z(b, q)$, $f(b, q, G)$, and $f(b, q, B)$ are all continuous in q for $b = 0.5$; the supremum over $q \in [0.5, 1]$ of $V(b, q)$ must therefore be attained, being that of a continuous function of q over a compact set. An optimal policy must accordingly exist at $b = 0.5$.

Consider now $b \in (0.5, p]$ and a stationary policy $\pi'(\cdot)$ with $\pi'(b) < b$. By Equation (3) and Equation (2), the application of such a policy leaves the public belief unaltered, induces every agent to select action G , and applies the same control at every future time step. Exploiting the facts that $b_i \leq p$, that $\mathbb{P}(a_i = G | b_i, \pi'(b_i)) = 1$, and that $b_{i+1} = b_i$, we may apply Equation (6) to write the total expected discounted utility under such a policy as follows:

$$V_B^{\pi'}(b_i) = -\beta(p - \pi'(b_i)) + \delta V_B^{\pi'}(b_i)$$

Thus the policy yields expected value $V_B^{\pi'}(b_i) = \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$. Note that, in this case, $V_B^{\pi'}(b_i)$ is an increasing function of $\pi'(b_i)$, so that the supremum of $V^*(\pi')_B(b_i)$ over $\pi'(b_i) \in (0.5, b)$ is not attained.

We now distinguish two cases: (1) $\max_{q_i \in [b_i, 1]} V(b_i, q_i) \geq \sup_{\pi'(b_i) \in (0.5, b_i)} \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$, and (2) its complement. In the former case, we may argue, as we did for $b = 0.5$, that the supremum must be attained and that an optimal policy exists with $\pi_B^*(b_i) \geq b_i$. In the latter case, as argued above, no optimal policy exists; instead, for sufficiently small $\epsilon > 0$, there exists an ϵ -optimal policy with $\pi_B^\epsilon(b_i) = b_i - \epsilon$. This yields claims (D) and (E).

It remains to show that there exists $t_2 \in (0.5, p)$ such that, for $b_i \in (t_2, p]$, we fall into the latter case, thereby yielding claim (F). We define functions $m(\cdot)$ and $n(\cdot)$ as follows, corresponding to the value resulting from a choice of precision below b or at least b , respectively:

$$\begin{aligned} m(b) &= \sup_{\pi'(b) \in [0.5, b]} \frac{-\beta(p - \pi'(b))}{1 - \delta} \\ n(b) &= \max_{q \geq b} V(b, q) \end{aligned}$$

Taking the limit of $m(\cdot)$ as $b \rightarrow p$ yields:

$$\lim_{b \rightarrow p} m(b) = 0 \tag{57}$$

When $q \geq b$ we are situated in the first case of Equation (49); we may thus expand $n(b)$ as follows:

$$\begin{aligned} n(b) &= \max_{q \geq b} [-\beta(|q - p|) - Cz(b, q) \\ &\quad + \delta y(b, q) V_B^*(b, q) \\ &\quad + \delta z(b, q) V_B^*(b, q)] \end{aligned} \tag{58}$$

We shall argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Since p is a fixed parameter strictly less than 1, this entails, by Equation (57), that $n(b) \leq -C(1 - p) < \lim_{b \rightarrow p} m(b)$, which in turn implies the existence of an open interval $(t_2, p]$ on which no optimal policy exists, thereby yielding claim (F). We now argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Observe that every term of Equation (58) is bounded above by 0, so that it suffices to show that some one term—say the second, $-Cz(b, q)$ —is bounded above by $-C(1 - p)$. Note first that $z(b, q) = q(1 - 2b) + b$, whence $z(b, q)$ is decreasing in q (since $b > 0.5$); thus $z(b, q) \geq z(b, 1) = 1 - b \geq 1 - p$, the final inequality following because $b \in (0.5, p]$. The claim follows since $p < 1$.

In conjunction with Equation (57), this implies the existence of an open interval (t_2, p) on which no optimal policy exists, thereby yielding claim (F).

Case 4: $b \in (p, 1]$ Claim (A)

For $b \in (p, 1]$, consider applying the stationary policy $\pi(\cdot)$ with $\pi(b) = p$. Since $b > p$, Equation (3) and Equation (2) imply that such a policy leaves the public belief unaltered, induces every agent to select action G , and applies the same control at every future time step. We may apply Equation (6) to write

$$V_B^\pi(b) = -\beta(p - p) + \delta V_B^\pi(b) = \delta V_B^\pi(b)$$

Thus $V_B^\pi(b) = 0$. Since this is likewise an upper bound of the optimal value function for any b , we conclude $\pi_B^*(b) = p$. ■

C.9 SOCIAL WELFARE MONOTONICITY

Lemma 17. Precision Monotonicity of Social Welfare

If $\pi_1(b) \leq \pi_2(b) \forall b \in [0, 1]$, for policies $\pi_1(\cdot), \pi_2(\cdot)$, then

$$W^{\pi_1}(b) \leq W^{\pi_2}(b) \forall b \in [0, 1]$$

Proof. It is manifest that the one-step welfare $-C \min(b_i, 1 - b_i, \pi(b_i))$ is increasing in $\pi(b_i)$.

Moreover, the convexity of social welfare with respect to the public belief follows from the very argument employed in the proof of Theorem 2, inasmuch as social welfare coincides with the altruistic value function in the special case $\beta(\cdot) = 0$.

We may then reprise the argument of the proof of Lemma 9 (Appendix C.4) to establish that an increase in precision augments welfare. $\square$

D EXTENSION TO HETEROGENEOUS AGENT PREFERENCES

Here we argue that, for the altruistic planner, the problem we have considered is equivalent to one comprising two types of individuals endowed with differing preferences.

Suppose there are two types of agents, G and B , the type of agent i being denoted t_i . The first type receives unit reward when $a_i = \omega$ and no reward otherwise; the second receives unit reward when $a_i \neq \omega$ and no reward otherwise. Suppose, further, that every agent knows both her own type and the types of all other agents. All remaining aspects of the model are unchanged.

The private belief update of agent i may be expressed as below. It remains unchanged, since it depends solely upon the signal structure, i.e., $\mathbb{P}(s_i = \omega) = q_i$.

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q}{1+2b_i q_i - b_i - q_i} b_i & s_i = G \\ \frac{1-q}{b_i + q_i - 2b_i q_i} b_i & s_i = B \end{cases} \quad (59)$$

By means of Equation (59), agent i 's action may be written as follows, the superscript indicating the type of agent i and $!$ denoting negation:

$$a_i^G = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & b_i < 1 - q_i \end{cases}$$

$$a_i^B = \begin{cases} !s_i & 1 - q_i \leq b_i \leq q_i \\ B & q_i < b_i \\ G & b_i < 1 - q_i \end{cases}$$

Since we assume each agent's type to be common knowledge, whenever $q_i \geq \max(b_i, 1 - b_i)$ all agents can infer agent i 's private signal exactly; whenever $q_i < \max(b_i, 1 - b_i)$, agent i 's action is uninformative.

Consequently, upon observing agent i 's type t_i , precision q_i , and action a_i , the public belief is updated as follows:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (60)$$

In this setting, an altruistic planner seeks to induce type- G agents to take action ω and type- B agents to take action $!\omega$, since these choices are concordant with the respective personal utility functions of the agents.

We may thus express the planner's instantaneous reward as follows, the superscript denoting the type of agent i :

$$r^G(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i \neq \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

$$r^B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

These are identical to one another and coincide with the reward function in Equation (5), whence follows the equivalence of the altruistic planner's problem in the presence of heterogeneous but observable types. $\blacksquare$

E EXPERIMENTS

E.1 EXPERIMENTAL SETUP

To simulate the controlled social learning setting, we consider a sequence of agents deliberating over whether or not to purchase a new model of car, informed by observations of past actions and by an advertisement whose precision is selected by the planner. LLMs discharged three distinct roles: (1) the agents, (2) the planner, and (3) an oracle. Each agent was assigned a ‘profile’ comprising demographic information together with her car use cases and preferences. At the outset, we designate either the good or the bad car as the offering. The information sets and objectives of the planner and the agents were exactly as in the model of Section 3.

The oracle’s role was twofold: (1) it generated a generic message to serve as the baseline-precision signal in the absence of any intervention by the planner; and (2) once the planner had chosen a precision, it generated a message of the prescribed precision, customized to the profile of the target agent.

The oracle was furnished with a fact sheet (safety rating, customer satisfaction, and the like) for each type of car, along with the agents’ profiles. Together, the fact sheet and the profile facilitated the generation of a tailored message of the precision prescribed by the planner. Further detail on the experimental setup and prompting may be found in Appendix E.2.

Each agent was charged with rendering a binary decision on the basis of observations of previous decisions and her private signal, the agent’s goal being to match the true parameter value.

The planner was furnished with the same decision history and charged with selecting a precision level for each agent’s message. The planner was additionally provided with the precision of a generic, untailored message that could be dispatched free of charge. In the altruistic case the planner’s objective was to induce correct decisions, whereas in the biased case its goal was to induce agents to purchase the car.

We compare the LLM simulations against the analytic solutions obtained via JuliaPOMDP (Egorov et al., 2017).

E.2 PARAMETERS AND PROMPTS

E.2.1 INSTANCE PARAMETERS

The experiments were conducted using the LG AI EXAONE Deep 32B model (LG AI Research, 2025). We solve the model analytically and run the LLM simulations for the parameters enumerated in Table 1.

Parameter	Description	Values
C	Cost of a bad action	1
k	Scaling factor for control cost	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$
p	Baseline signal precision	$\{0.6, 0.7, 0.8, 0.9\}$
δ	Discount Factor	$\{0, 0.25, 0.5, 0.75, 1\}$

Table 1: Instance parameters employed in the analytical and LLM-based experiments.

The car facts were contrived so that the good car was good for everyone and the bad car was bad for everyone. Below we reproduce the fact sheets for the GOOD and BAD cars, respectively.

```
fact_sheet["GOOD"] = [  
    "EPA-rated range: 290 miles.",  
    "Fast-charge 10-80 %% in 24 min on 150 kW DC.",  
    "8-year / 100,000-mile battery warranty.",  
    "NHTSA 5-Star overall safety rating.",  
    "Cold-soak -5 °F keeps 93 %% range.",  
    "Cools cabin 100→72 °F in 3 min.",  
    "Cargo volume 26 ft3 total.",  
    "Tow's 1 500 lb with factory hitch.",  
    "Average 1.2 service visits first 3 yrs.",  
    "92 %% owners would buy again (Apr-25).",  
    "Qualifies for federal credit.",  
    "Quarterly OTA updates incl. battery pre-heat."  
]  
fact_sheet["BAD"] = [  
    "EPA-rated range: 155 mi (combined).",
```

```

1890         "Fast-charge 10 %% → 80 %% in 51 min (50 kW DC hardware limit).",
1891         "Battery warranty: 3 yr / 36 000 mi, first owner only.",
1892         "NHTSA overall safety rating 3 Stars (frontal-offset weakness).",
1893         "Cold-soak 15 °F retains 58 %% range; cabin heat cuts range 20%.",
1894         "Cabin cools 105 °F → 72 °F in 11 min.",
1895         "Cargo volume 17 ft³; rear seats fixed (no fold-flat).",
1896         "No factory towing approval.",
1897         "Average 3.8 service visits per year (battery-alert issues).",
1898         'Owner survey: 41 %% "would buy again" (1 200 owners, Feb-25).',
1899         "Not eligible for federal tax credit.",
1900     ]

```

The profiles were generated programmatically, each encompassing an agent's name, age, city of residence, interests, intended use case for the car (e.g., commuting, road trips, and the like), and principal pain points or concerns (e.g., fuel efficiency, winter driving, and the like).

E.2.2 PROMPTS

Agent LLM Prompt:

```

1908 sys = (
1909     f"You are deciding whether to buy a car. "
1910     f"Your profile is {profile}"
1911     f"You start believing there's a 50% chance the car is GOOD. "
1912     f"You have two information sources: "
1913     f"1. Previous agents' decisions "
1914     f"2. An advisor's message about the car "
1915     f"Your goal is to maximize your reward "
1916     f"i.e. BUY if the car is GOOD and "
1917     f"NOT BUY if the car is BAD. "
1918     f"If you BUY a GOOD car, you GAIN ${C}. "
1919     f"If you NOT BUY a BAD car, you GAIN ${C}. "
1920     f"If you BUY a BAD car, you LOSE ${C}. "
1921     f"If you NOT BUY a GOOD car, you LOSE ${C}. "
1922     f"belief = your probability (0.0-1.0) that the car is GOOD. "
1923     f"decision = exactly 'BUY' or 'NOT BUY'."
1924 )
1925 user = (
1926     f"PREVIOUS DECISIONS:\n{history_txt or '(none)'}\n\n"
1927     f"ADVISOR MESSAGE:\n{message}\n\n"
1928     f"TASK: Use the previous decisions and the advisor message. "
1929     f"Decide BUY or NOT BUY. "
1930     f"Update your belief about car quality (0.0-1.0). "\n"
1931     f'JSON response:{{"belief": <number>,"decision": "<BUY or NOT BUY>"}}'
1932 )

```

Planner LLM Prompt:

```

1931 sys = (
1932     f"You are providing messages to potential car buyers. "
1933     f"Each potential car buyer (or agent) has the same information. "
1934     f"Your goal is to get agents to {objective}."
1935     f"Each decision according to your objective, earns you {C} dollars. "
1936     f"You and the agents have the same information. "
1937     f"Initially there is a 50% chance the car is GOOD. "
1938     f"The agents' aim to BUY if the car is GOOD "
1939     f"and NOT BUY if the car is BAD. "
1940     f"You can send a baseline message for free "
1941     f"or tailor the message to the agent at a cost. "
1942     f"Tailoring costs {k} dollars times the change in precision. "
1943     f"(Message precision is the probability (0.5 to 1.0) "
1944     f"that the message leaves the buyer with the correct impression. "
1945     f"For example, if the car is likely GOOD: "
1946     f"a high precision message will likely suggest BUY "

```

```

1944     f"and a low precision message will likely suggest NOT BUY. "
1945     f"You may choose to increase or decrease precision. "
1946     f"Agents decide based on your message and the decision history. "
1947     f"They update beliefs by observing previous decisions. "
1948     f"IMPORTANT: Thus your objective is to maximize:"
1949     f"{C} * (# of agents who choose according to your objective) "
1950     f"- sum of ({d}^i * {k} * abs(message_precision - baseline_precision)) \n"
1951     f"where {d} is your discount factor for future cost. "
1952 )
1953 user = (
1954     f"BASELINE MESSAGE precision:\n{pi_base:.2f}\n"
1955     f"DECISION HISTORY:\n{history_txt}\n"
1956     f"TASK:"
1957     f"Based on the history and your objective, "
1958     f"choose your desired precision.\n"
1959     f'JSON response: {"message_precision": <number>}\n'
1960 )

```

Oracle LLM Prompt:

```

1961 sys = (
1962     f"You will be given "
1963     f"1. an agent profile, "
1964     f"2. information about a car, "
1965     f"3. the true quality of the car, "
1966     f"4. a baseline message and the 'precision' of that message, and "
1967     f"5. a desired precision for the message you output. "
1968     f"(Message precision is the probability (0.5 to 1.0) "
1969     f"that the message leaves the buyer with the correct impression. "
1970     f"For example, if the car is likely GOOD: "
1971     f"a high precision message will likely suggest BUY "
1972     f"and a low precision message will likely suggest NOT BUY. "
1973     f"IMPORTANT: Output a message with the requested precision. "
1974     f"IMPORTANT: The message should be tailored to the profile. "
1975 )
1976 user = (
1977     f"AGENT PROFILE:{profile}\n"
1978     f"TRUE CAR QUALITY: {omega}\n\n"
1979     f"CAR FACTS: {car_facts}\n"
1980     f"BASELINE MESSAGE:{baseline_message}\n"
1981     f"BASELINE MESSAGE precision: {p}\n"
1982     f"DESIRED MESSAGE precision: {q}\n"
1983     f"IMPORTANT: Output a message with the requested precision. "
1984     f"TASK: Write a message (MAX 100 words) with the desired precision."
1985     f'JSON response: {"message": "<message>"}\n'
1986 )

```

E.3 VALIDATION OF LLM BELIEFS AND ORACLE

We now adduce further experiments to validate the self-reported beliefs of the LLMs (as displayed in figure 1b) and the performance of the LLM oracle. In these experiments, we treat the LLMs as rational, utility-maximizing—though not necessarily Bayesian—agents. The aim of these experiments is twofold: (1) to demonstrate that the model’s self-reported posterior genuinely corresponds to the belief that governs its choices in state-contingent decision problems; and (2) to verify that the messages produced by the oracle do indeed possess the desired precision.

First, we furnish the LLM with a prior and a new signal, elicit its self-reported posterior belief, and then request that it render a binary “buy/not-buy” decision yielding unit reward if and only if the true state is good. A rational agent ought to choose “buy” precisely when its subjective belief exceeds 0.5. As figure 4a shows, the LLM’s decisions accord sharply with this threshold: it chooses “buy” overwhelmingly when its reported posterior exceeds 0.5 and “not buy” otherwise. Next, we elicit the model’s willingness to pay for a payoff contingent upon the true state. For a range of prices c , we inquire whether the agent would pay c in exchange for receiving unit utility should the state be good. A rational agent with belief p ought to accept precisely when $c < p$. Figure 4b plots the fraction of “buy” decisions across posterior–price pairs, revealing a clear separation: acceptance rates are high when $c < p$ and low when $c > p$, exactly as one would anticipate from a

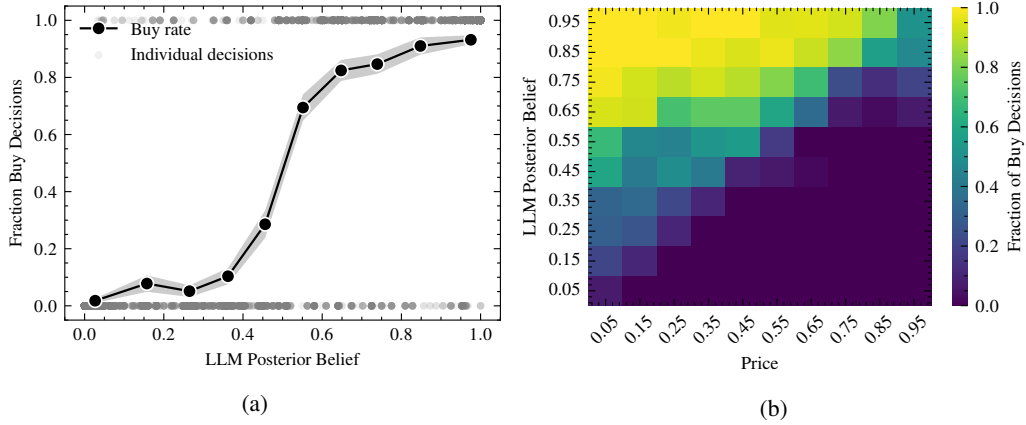


Figure 4: The rate at which the LLM elected a ‘buy’ decision, as a function of (a) its self-reported posterior belief and (b) its self-reported posterior together with the price of buying.

utility-maximizing decision maker whose belief is p . Both experiments thus demonstrate that the self-reported posterior lies close to the model’s inherent belief as deployed in decision making.

In similar fashion, we validate the oracle’s performance in crafting messages of a given precision. After the oracle has generated a message of specified precision q , we request that a separate LLM form an estimate $\hat{q}$ of the message’s precision. We do so for $q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ over 100 programmatically generated agent profiles, with 10 trials per combination, amounting to 6,000 trials in total. The mean absolute error $|q - \hat{q}|$ was a mere 0.025, which indicates that the realized precision of the oracle’s messages closely tracks the requested precision.

E.4 IMPACT OF NON-BAYESIAN LLM BELIEF UPDATES ON LEARNING TRAJECTORIES

We now unpack the belief trajectories arising from an example instance of the LLM experiment, so as to elucidate the impact of non-Bayesian belief updating. Here we fix the policy to be the analytically optimal one and apply it to LLM-simulated agents. When the car is bad (right panel), the altruistic planner drives the Bayesian belief (dotted black line) rapidly toward 0; the LLM belief (solid black line), by contrast, proves more “stubborn.” This macro-level trajectory is a direct consequence of the micro-level biases: the public belief resists entering a cascade (NB3) because individual agents overreact to the occasional positive signal that runs counter to the group’s increasingly negative belief (NB2).

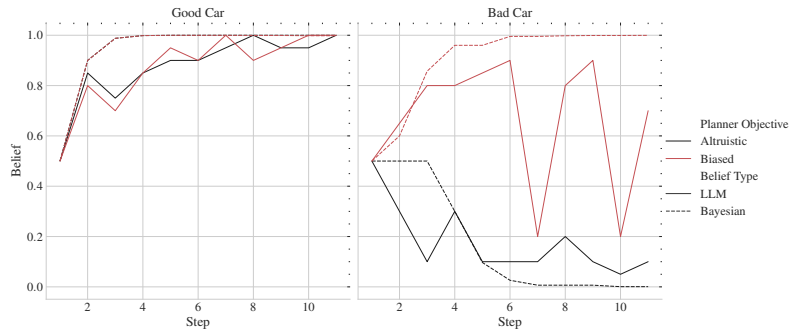


Figure 5: Agent belief-updating trajectories for the true Bayesian belief and for the belief reported by the LLM agent, under analytically optimal policies, across differing planner objectives and true states of the world.